

CHROMATOGRAPHY

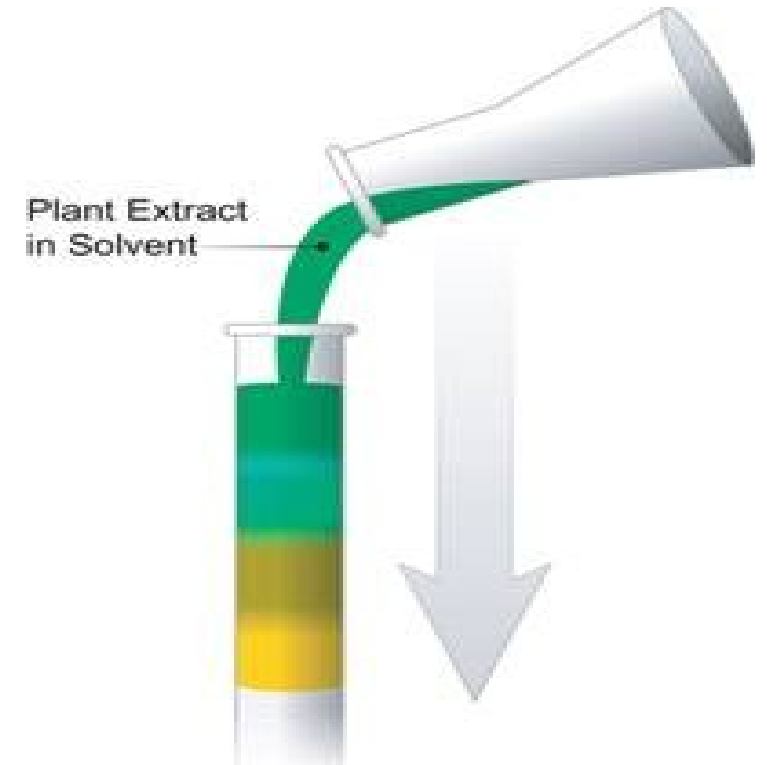
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History of Separation Techniques

- Mikhail Semyonovich Tsvet ,a Russian botanist, in 1903 first coined the term chromatography.
- during his research on plant pigments.
- He used liquid-adsorption column chromatography with calcium carbonate as adsorbent and petrol ether/ethanol mixtures as eluent to separate chlorophylls and carotenoids .

Tall glass open column filled with sand like particles.

- Ground up plant extract poured into the column
- Colored bands develop as the extract percolated down through the column.
- Different compounds had separated.



Introduction to Separation Techniques

What is Chromatography?

□ Greek word

- Chroma- color Graphy- Writing

Color writing

- **Chromatography** is a separation process that is achieved by distributing the components of a mixture between two phases, a **stationary phase** and a **mobile phase**.
- Those components held preferentially in the stationary phase are retained longer in the system than those that are distributed selectively in the mobile phase.
- Therefore, chromatographic separations are a consequence of **differential migration** of solutes.

- **analyze, identify, purify, and/or quantify** the mixture or components.
- **Mobile phase** – is a moving phase that continuously flows through the stationary phase and carries the analyte.
- **Stationary phase** –the phase that is either a solid or liquid particle attached to a glass or a metal surface on which the components of the mixture to be separated are absorbed selectively.
- In a chromatographic separation of any type, **different components of a sample are transported in a mobile phase** (a gas, a liquid, or a supercritical fluid).
- The mobile phase (also called **eluent**) penetrates or passes through a solid or immiscible stationary phase.
- Solutes (**eluates**) in the sample usually **have differential partitioning or interactions** with the mobile and stationary phases.
 - ***This results in differential migration***

Components are separated based on :

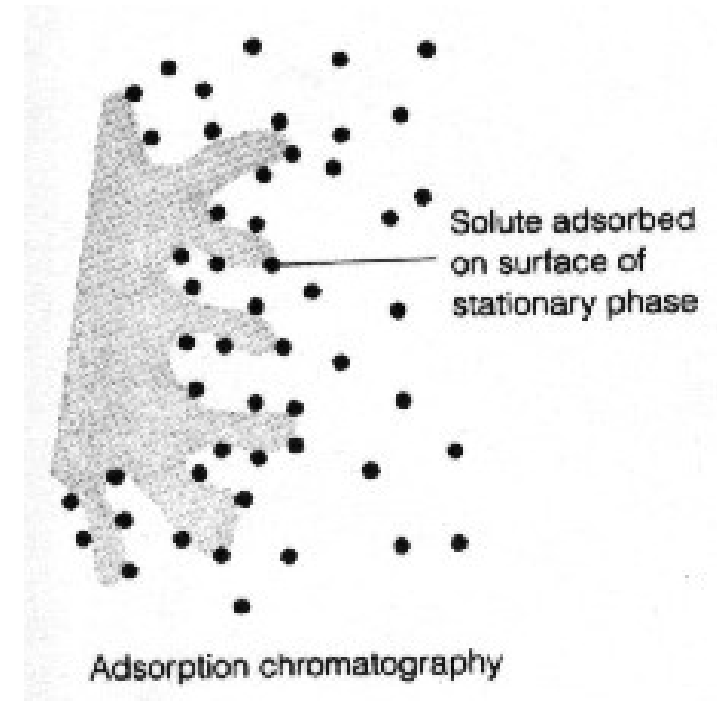
- polarity
- boiling point
- ionic strength
- size

There are different types of chromatography.

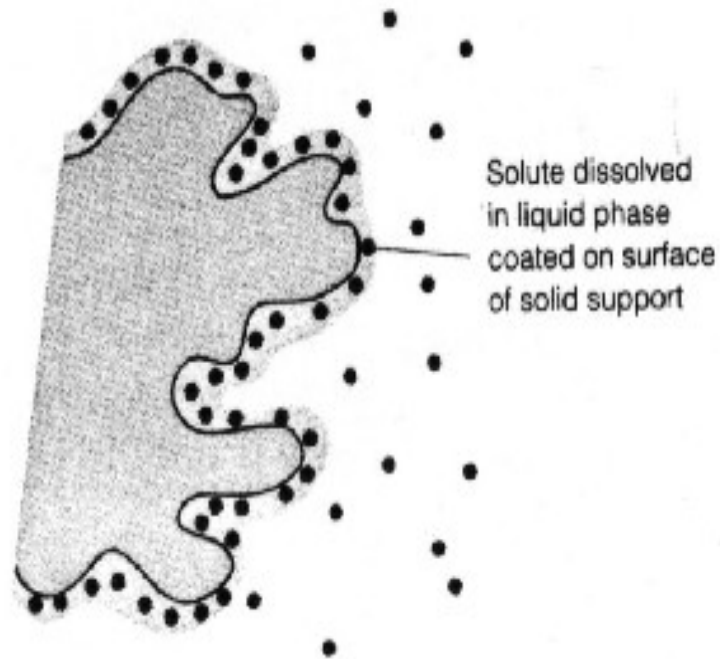
Types of chromatography (according to distribution between mobile phase and stationary phase.)

1. Adsorption Chromatography

- It depends on the ability of different solutes to be adsorbed on the surface of the stationary phase at different strengths
- It can be found in GLC & LSC.
- **Solid SP and Liquid/gas MP**



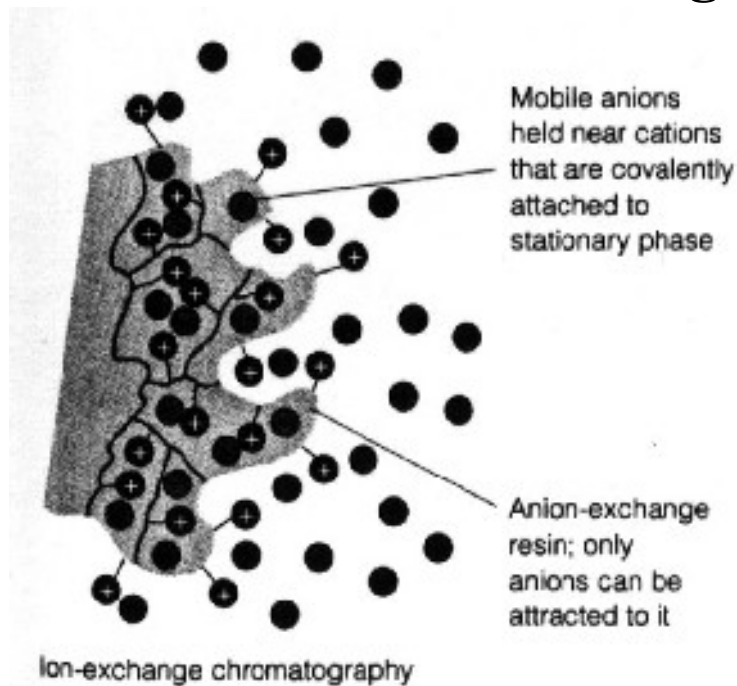
Partition Chromatography



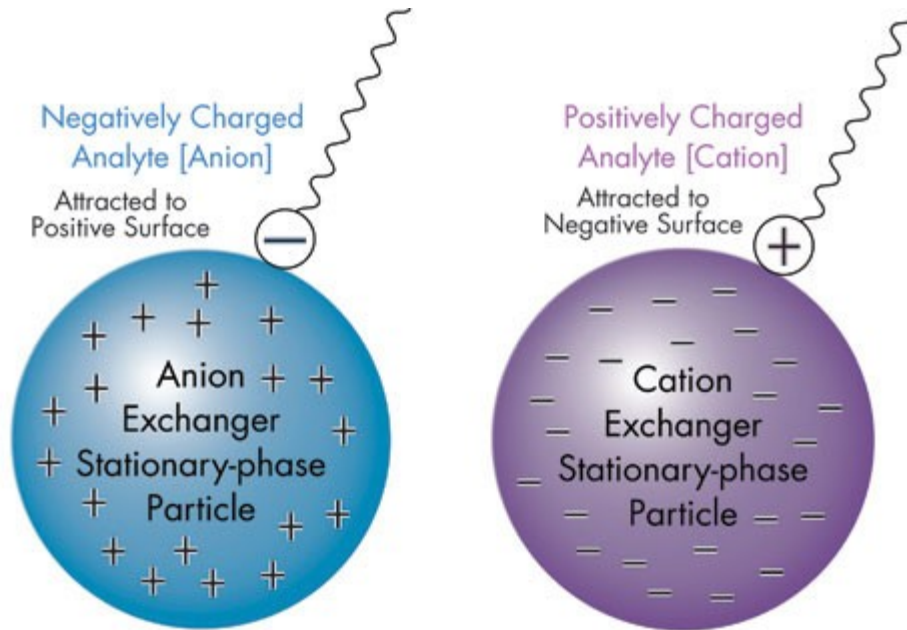
Partition chromatography

- **Liquid SP and Liquid/gas MP**
- Molecules will partition into the stationary phase based upon affinity for stationary phase & eventually partition into mobile phase again
- Thin layer of the liquid SP is coated onto inside of GC column or on small particles on LC column

Ion Exchange Chromatography



- A solid polymeric stationary phase containing replaceable ions
- Ionic liquid mobile phase
- Separation of either cations or anions
- Separation based on relative strength of ionic bond.
- Solute ions of the opposite charge in the mobile liquid phase are attached to the resin by electrostatic force.
- Anion exchange has cations ((e.g., $\text{N}(\text{CH}_3)^{3+}$ on surface
- Cation exchange has anions (eg. SO_3^-) on the surface



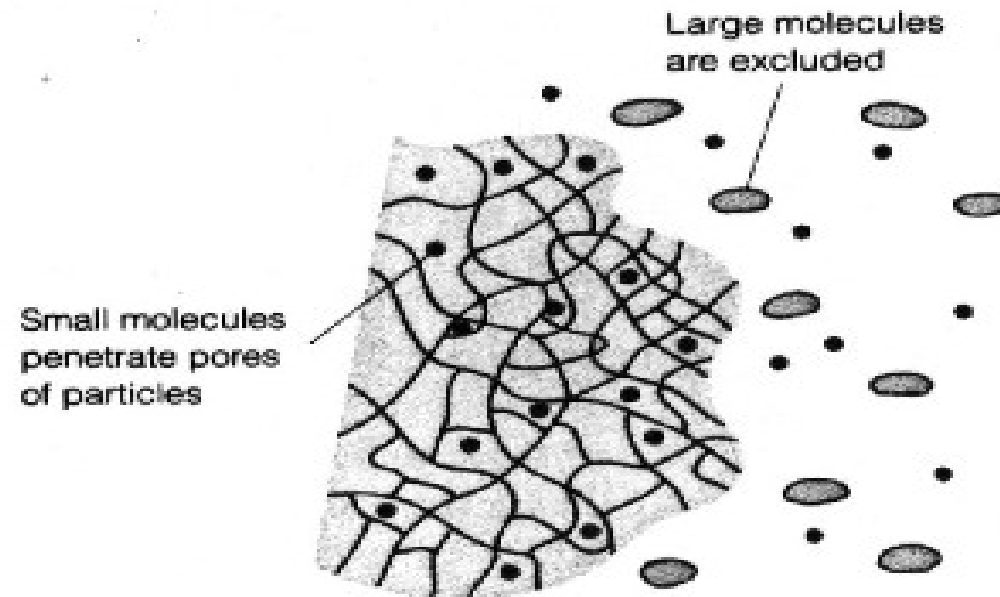
- the use of a resin (the stationary solid phase) is used to covalently attach anions or cations onto it. Solute ions of the opposite charge in the mobile liquid phase are attracted to the resin by electrostatic forces.

Molecular Exclusion Chromatography

- Also known as **gel permeation** or **gel filtration**.
- Lacks an attractive interaction between the stationary phase and solute.
- The liquid or gaseous phase passes through a porous gel which separates the molecules according to their **size**.
- The pores are normally small and exclude the larger solute molecules, but allows smaller molecules to enter the gel, causing them to flow through a larger volume.
- This causes the larger molecules to pass through the column at a faster rate than the smaller ones.

Molecular Exclusion Chromatography...

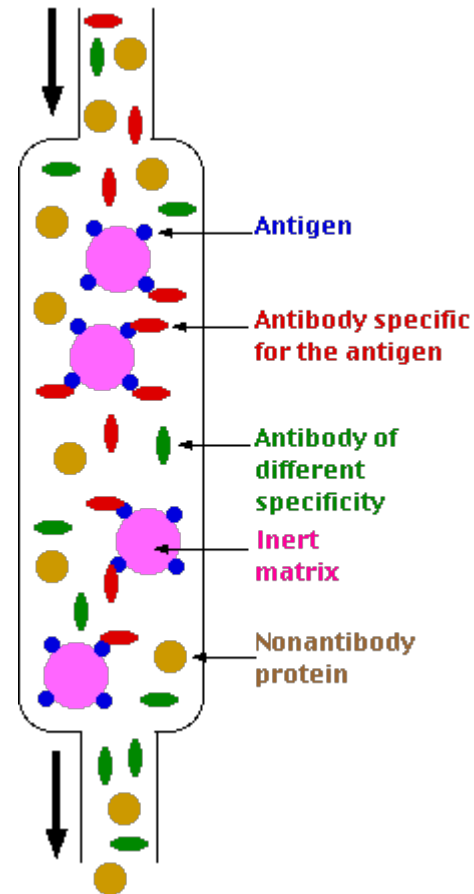
- Small molecules get trapped in pores & take longer to get out



Molecular exclusion chromatography

Affinity Chromatography

- **Affinity chromatography** is a method of separating biochemical mixtures based on a highly specific interaction such as that between antigen and antibody, enzyme and substrate, or receptor and ligand



Column Chromatography

- is a separation technique in which the stationary bed is within a tube.
- The particles of the solid stationary phase or the support coated with a liquid stationary phase may fill the whole inside volume of the tube (**packed column**)
- or be concentrated on or along the inside tube wall leaving an open, unrestricted path for the mobile phase in the middle part of the tube (**open tubular column**).
- Differences in rates of movement through the medium are calculated to different retention times of the sample .

Column Chromatography...

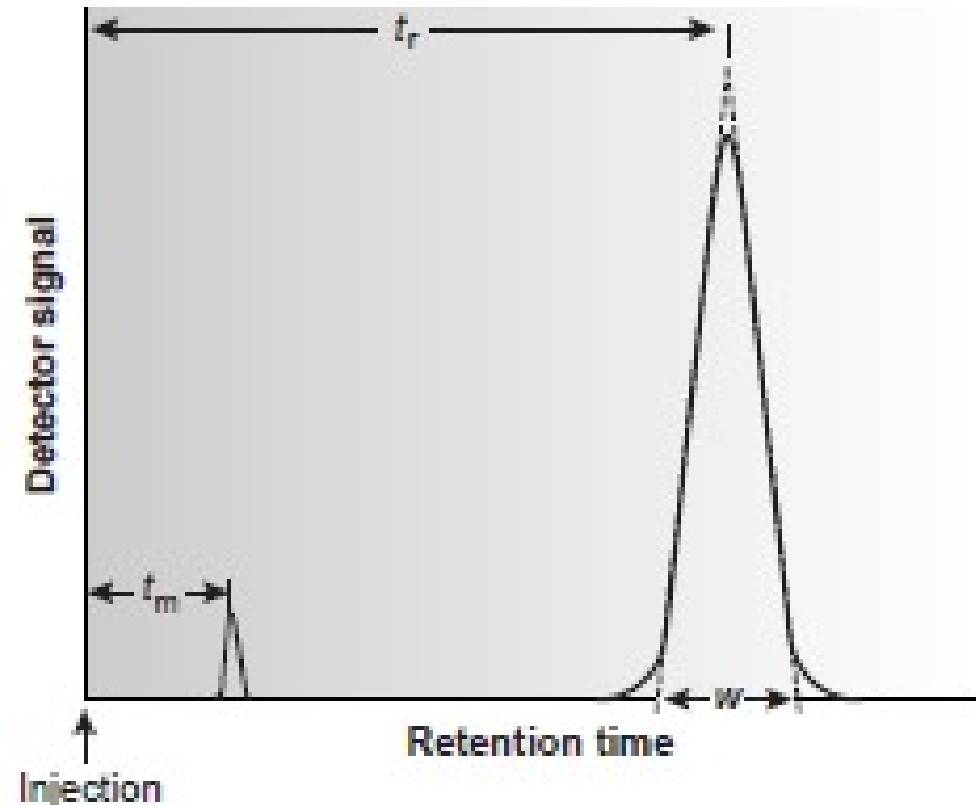
- The progress of a chromatographic separation is monitored with a suitable **detector** situated at the end of the column.
- A plot of the detector's signal as a function of time or volume of eluted mobile phase is known as **chromatogram** and consists of a peak for each of the separated solute bands.

Column Chromatography...

- The chromatogram can be characterized

By its **retention time** and **peak width**

- The time taken for the mobile phase to pass through the column is **called t_M**



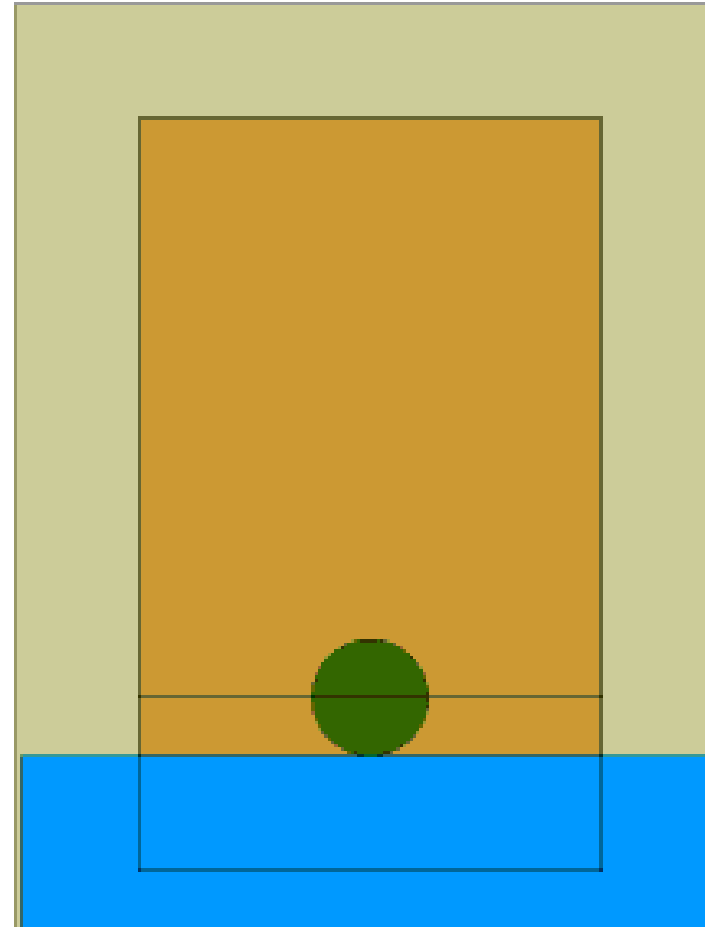
Planar Chromatography

- is a separation technique in which the stationary phase is present as or on a plane.
- The plane can be a paper, serving as such or impregnated by a substance as the stationary bed (paper chromatography) or a layer of solid particles spread on a support such as a glass plate (thin layer chromatography).
- Different compounds in the sample mixture travel different distances according to how strongly they interact with the stationary phase as compared to the mobile phase.
- The specific Retention factor (R_f) of each chemical can be used to aid in the identification of an unknown substance.

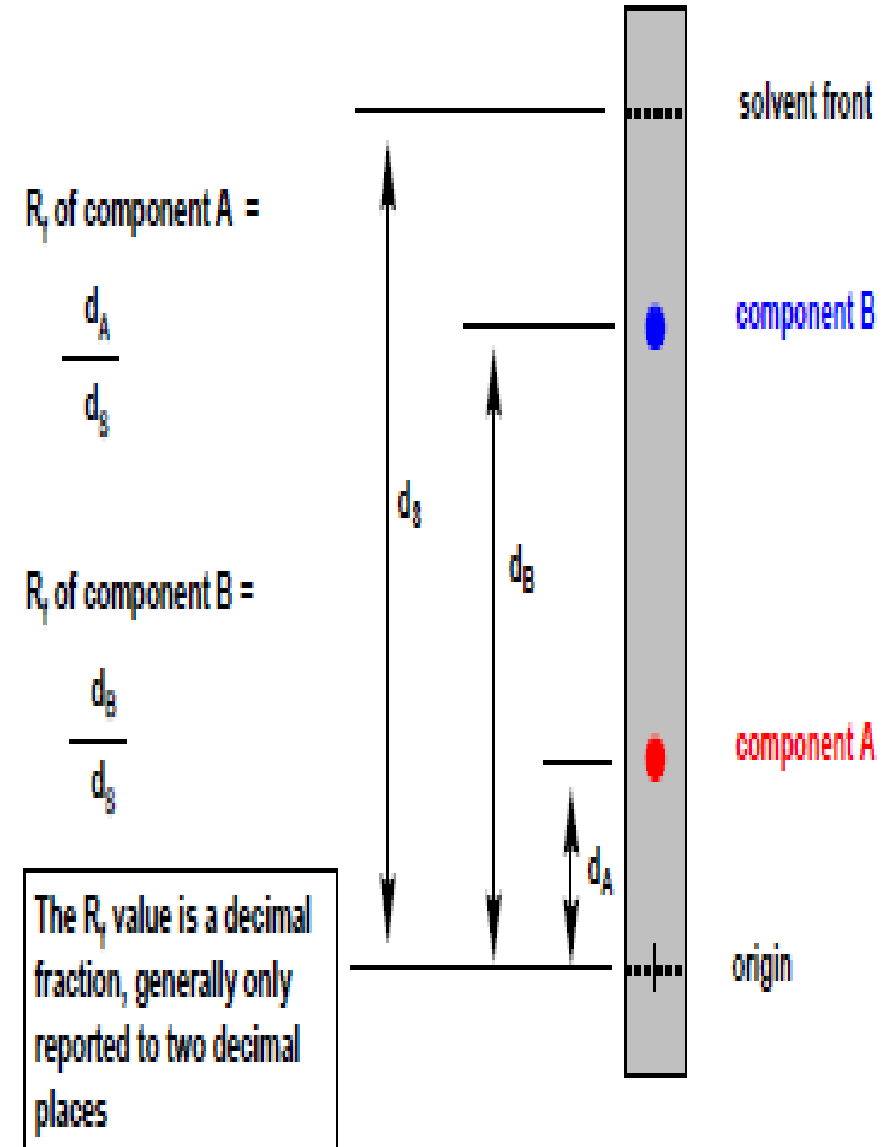
Planar Chromatography...

Put small amount of the Mobile phase into Chromatographic tank

- ✓ Saturate it
- ✓ Take Paper or TLC coated plate
- ✓ Draw two lines (bottom and upper)
- ✓ Apply sample on the bottom line as spot or band
- ✓ Develop
- ✓ Drying and detection



- Detection of separated compounds in PC and TLC is made either
 - By its natural color
 - By scanning under UV fluorescence
 - By spraying with reagents
- Characterization of the compounds is made by measuring retardation factor (R_f)



Principle of planar chromatography

- is based on partition chromatography
- adsorption chromatography can take place where the stationary phase is the solid surface of the paper and the mobile phase is the liquid phase.

Paper Chromatography Procedure

1. Selection of the ideal type of development: Based on factors such as the complexity of the solvent, mixture, paper, etc. the development type is chosen.

❑ Mostly either **Radial or Ascending** type of paper chromatography is employed.

✓ because of the easiness they offer while handling and performing

✓ obtaining the chromatogram faster within a shorter duration of time.

2. Selection of paper: As per the pores' size and the sample quality.

2. Sample preparation: This involves the dissolution of the sample in an ideal solvent that is being utilized in developing the mobile phase.

4.Sample loading or spotting on the paper: With the help of a capillary tube or a micropipette, the sample is spotted on the paper at an accurate position.

- This promotes the interpretation of the chromatogram more quickly and easily.

5.Chromatogram development: This is carried by the paper immersion in the mobile phase. The mobile phase crosses over the sample on the paper because of the capillary action of the paper.

6.Drying of paper and detection of the compound: With the aid of air drier, the paper is dried as soon as the chromatogram is developed.

- On the chromatogram developed paper, the detecting solution is sprayed and dried thoroughly for the identification of the sample chromatogram spots.

Types of paper Chromatography

- **Ascending Paper Chromatography**
the developing solvent is found to be moving in **an upward direction**. Here, a sufficient quantity of mobile phase is poured into the development chamber.
- Sample and reference are spotted on the line drawn a few centimeters from the bottom edge of the paper suspended from a hook or clip at the top.
- **Descending Paper Chromatography**
the solvent front travels down the length of paper suspended from the top inside the developing chamber.
- The mobile phase is kept in a trough in the upper chamber. The paper with spotting on the line drawn a few centimeters from the top is clamped to the top. Before elution, the jar is covered and equilibrated with the mobile phase vapor.

- **Planar Chromatography**

SP-plane support

- **Paper chromatography**
 - **Liquid SP-soaked in cellulose paper**
- **Thin layer chromatography**
 - **adsorbent (Al_2O_3 or SiO_2 , usually) coating a sheet of plastic or glass**

Separated compounds-
appear as spot

- **Column chromatography**

- **SP-in column**

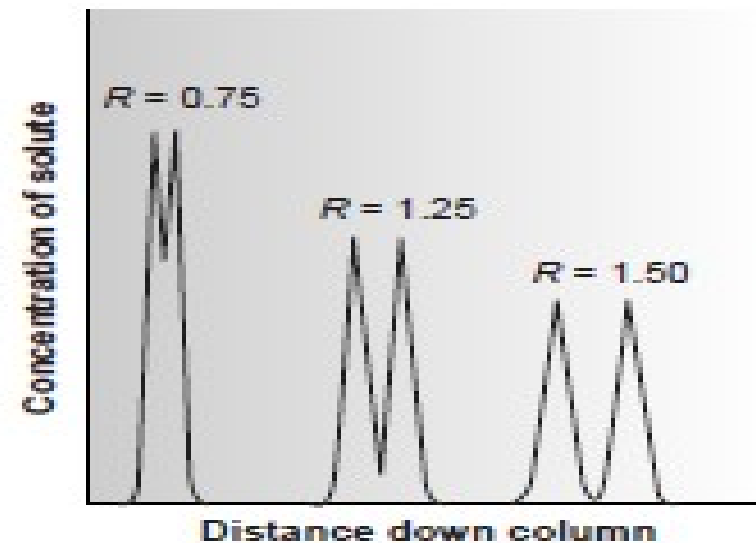
- **Gas chromatography**
- **High performance liquid chromatography**

- **Separated compounds-
appear as a peak**

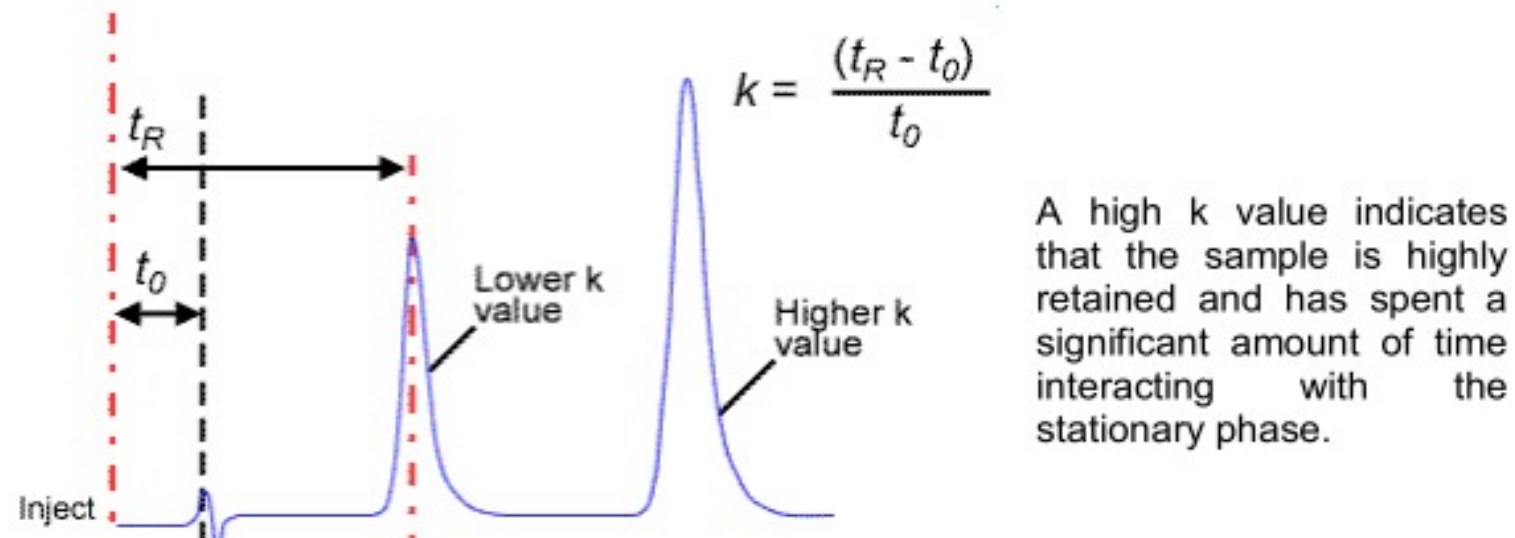
Chromatographic Resolution

- The goal of chromatography is to separate a sample into a series of chromatographic peaks, each representing a single component of the sample.
- **Resolution is a** quantitative measure of the degree of separation between two chromatographic peaks, A and B, and is defined as

$$R = \frac{t_{r,B} - t_{r,A}}{0.5(w_B + w_A)} = \frac{2\Delta t_r}{w_B + w_A}$$

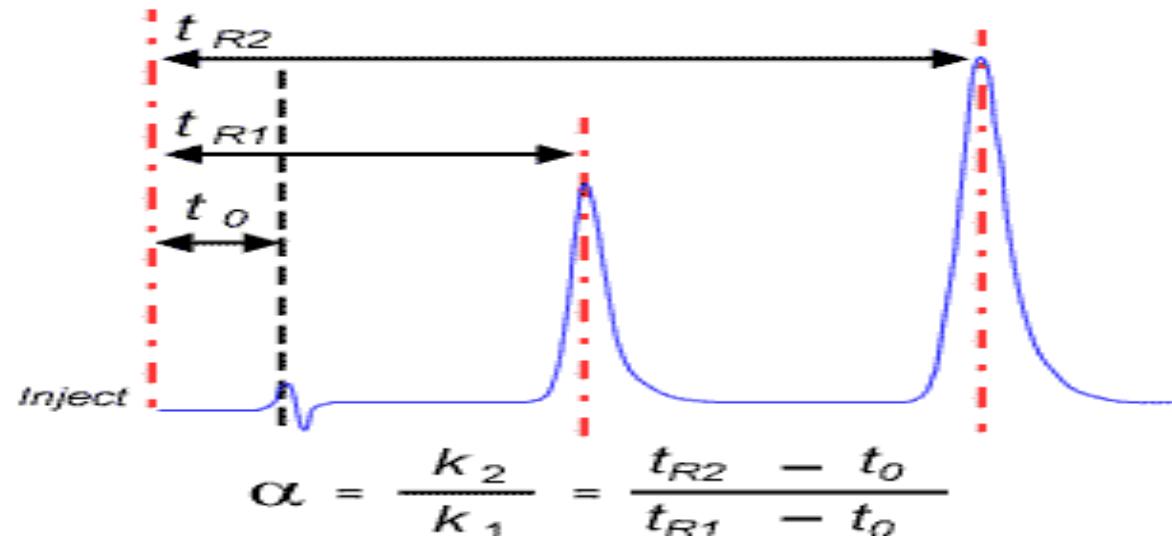


- is often used to describe the **migration rate of an analyte** on a column
- When an analytes retention factor **is less than one**, elution is so fast that accurate determination of the retention time is very difficult.
- High retention factors (greater than 20) mean that elution takes a very long time. (Ideal---1 to 5)



Column Selectivity

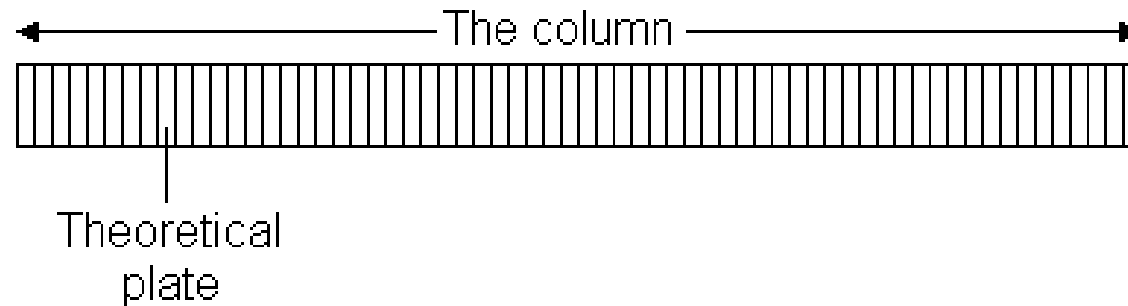
selectivity factor, α is the ratio of the amount of time the later eluting analyte spends in the stationary phase relative to that of the earlier eluting analyte.



- The selectivity factor is always greater than one (species A elutes faster than species B).

Column efficiency

- Separation power of the column
 - Efficiency is determined by the number of **theoretical plates**
 - Separate equilibrations of the sample between the stationary and mobile phase occur in these "**plates**"



$$N = \frac{L}{H}$$

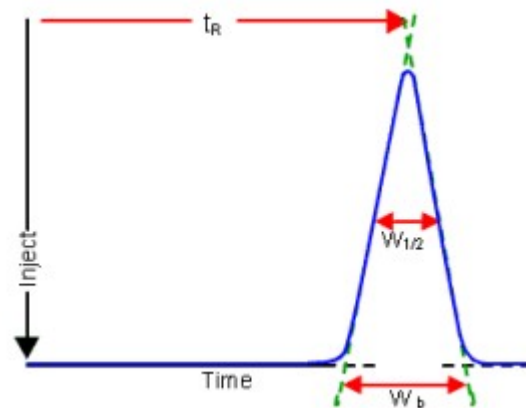
- Column efficiency

- The number of theoretical plates that a real column possesses can be found by examining a chromatographic peak after elution;

$$N = 16 \left(\frac{t_r}{w} \right)^2$$

$$N = 5.545 \left(\frac{t_r}{w_{1/2}} \right)^2$$

where $w_{1/2}$ is the peak width at half-height.



- Resolution can be increased by increasing
 - Capacity factor
 - Decrease in T^0 in GC
 - Decrease solvent strength in LC
 - Increasing the volume of stationary phase
 - Selectivity factor – by changing the S_p and M_p
 - Number of theoretical plates-decreasing particle size of the stationary phase

Types of Chromatography according to the kind of Mobile & Stationary Phases

1. Gas-Liquid Chromatography (GLC).
- 2- Gas-Solid Chromatography (GSC).
- 3- Liquid-Liquid Chromatography (LLC).
- 4- Liquid-Solid Chromatography (LSC).

GAS CHROMATOGRAPHY

Introduction

- GC was originally developed in the 1950s and commercialized in the late 1950s.
- It is a process of separating component(s) from the given crude drug by using a gaseous mobile phase.
- It involves a sample being vaporized and injected onto the head of the chromatographic column.
- The sample is transported through the column by the flow of inert, gaseous mobile phase. The column itself contains a liquid stationary phase which is adsorbed onto the surface of an inert solid.
- solute's retention is determined mostly by its *vapor pressure* and *volatility* and by *its interaction with the stationary phase*.

❖ Two major types

✓ **Gas-solid chromatography (stationary phase: solid)**

The mobile phase is a gas while the stationary phase is a solid. Used for separation of low molecular gases, e.g., air components, H_2S , CS_2 , CO_2 , rare gases, CO and oxides of nitrogen

- same material is used as both the stationary phase and support material
- common adsorbents include:
 - alumina
 - molecular sieves (**crystalline aluminosilicates [zeolites]**
and clay)
 - silica
 - active carbon
- long column lifetimes
- ability to retain and separate some compounds not easily resolved by other GC methods

✓ **Gas-liquid chromatography (stationary phase: immobilized liquid)**

The mobile phase is a gas while the stationary phase is a liquid retained on the surface as an inert solid by adsorption or chemical bonding.

- over 400 liquid stationary phases available for GLC
- material range from polymers (polysiloxanes (silicones), polyesters, polyethylene glycols) to fluorocarbons, and liquid crystals etc.
- Based on polarity, of the 400 phases available only 6-12 are needed for most separations.

Bonded-Phase Gas chromatography

- ▣ covalently attach stationary phase to the solid support material
- ▣ avoids column bleeding in GLC.
- ▣ bonded phases are prepared by reacting the desired phase with a silica-surface.
- reactions form an Si-O-Si bond between the stationary phase and support.

Principle

- The principle of separation in GC is “**partition.**”
- The mixture of component to be separated is converted to vapor and mixed with gaseous mobile phase.
- The component which is more soluble in stationary phase travel slower and eluted later.
- The component which is less soluble in stationary phase travels faster and eluted out first.
- No two components has same partition coefficient conditions.
- components are separated according to their partition coefficient.
- Partition coefficient is “the **ratio of solubility** of a substance distributed between two immiscible liquids at a constant temperature.’

Gas Chromatography...

- Most GC requires small amount of the sample and no attempt is made to collect the separated compounds.
 - **Analytical GC**
- However large columns are available which can separate samples up to ***one gram***.
- Receivers attached to the outside of the chromatography can collect the individual compounds after passing through the detectors.
 - **Preparative GC**

Gas Chromatography...

- **What Types of Compounds are Suitable for GC Analysis?**
 - For a compound to be suitable for GC analysis, it must possess appreciable **volatility** at temperatures below **350–400 °C**.
 - The compound must be able to **withstand high temperatures** and be rapidly transformed into a vapor without degradation or reacting with other compounds.
 - As a general rule, the **greater the molecular weight** or **polarity** of a compound, the lower is its volatility.
 - The presence of polar functionalities such **as hydroxyl and amine groups** severely decrease compound volatility (Eg. **Sugars and amino acids**)

Gas Chromatography...

- **What type of ...**
 - As a rule, **inorganic compounds** are not suitable for GC analysis. Metals and salts do not possess the required volatility.
 - Many **organo-metallics** have sufficient volatility for analysis due to the high organic content of these molecules.
 - However, there are many exceptions. Many **biomolecules and pharmaceuticals are thermally sensitive** and degrade at the temperatures used in gas chromatography
 - Overall, it has been estimated that only about 10% of all compounds can be analyzed by GC.

Gas Chromatography...

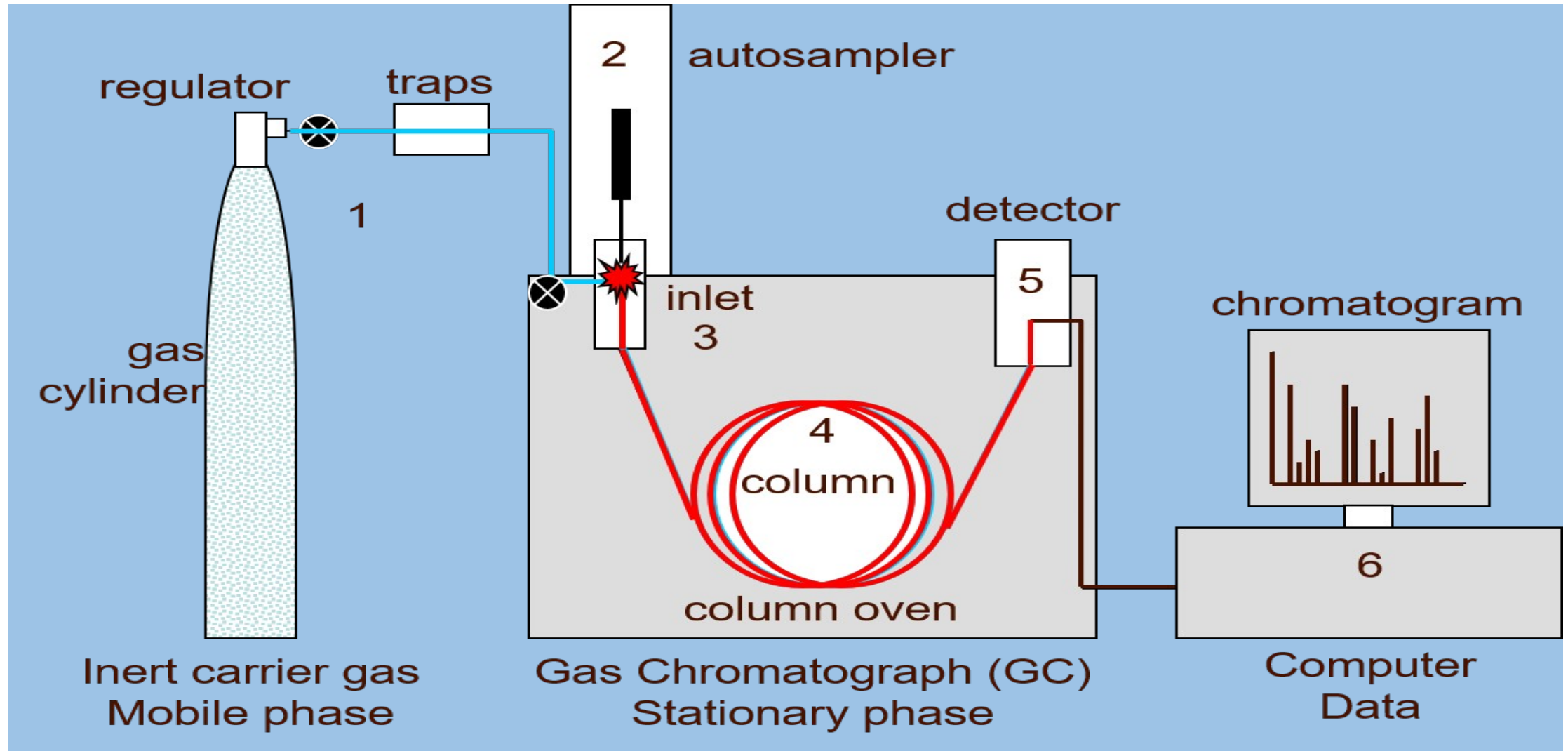
Advantages

- Gases diffuse at higher rate than liquids-
shorter analysis time
- Fast equilibration condition-**higher efficiency**
- Picogram (10^{-12}) amount of sample can be detected-**high sensitivity**
- Detectors specific for elements like **N, S & Cl**

Disadvantages

- Directly Applied to **volatile** compounds only
- Samples should not be **thermo-labile**
- **GC is expensive**

Instrumentation



Carrier Gas

The tank is fitted with:-

- pressure controller to control the pressure of gas,
 - pressure gauge that indicates the pressure
 - a molecular sieve to transfer filtered dry gas
 - a flow regulator to ensure a constant rate of flow of mobile phase to the column.
- ❖ It should meet the following criteria:
- ✓ Should be chemically inert.
 - ✓ Should be cheap and readily available.
 - ✓ Should be of high quality and not cause any fire accidents.
 - ✓ Should give best possible results.
 - ✓ Should be suitable for the sample to be analyzed and for the detector.

Carrier Gas...

- Hydrogen, helium, nitrogen and carbon dioxide are commonly used.
- **He, Ar, and N₂**, have the advantage of being chemically inert toward both the sample and the stationary phase.
- Hydrogen has low density and better thermal conductivity. However, it reacts with unsaturated compounds(hydrogenation) and is inflammable and explosive in nature.
- Nitrogen is inexpensive but it gives reduced sensitivity due to it's lower diffusion rate.
- **He** is the most preferred gas.

Inlet pressure ranges from: 10 -50 psi -Flow rate : 25 -150 mL/min for packed columns
-Flow rate: 2-25 mL/min for open tubular column

□ The choice of carrier gas is often dependent upon the type of detector which is used.

Injectors

- The injector introduces the sample into the open /packed/ tubular column.
- The injector is a hollow, metal cylinder containing a glass liner or insert.
- The column is inserted into the bottom of the injector so that the column end resides in the lower region of the glass liner.
- A liquid, or sometimes a gas, is introduced into the injector through a **resealable septum** using a small syringe

Con't...

The injector is heated to 100–300 °C, thus any volatile sample components are rapidly transformed into a vapor.

- The carrier gas mixes with the vaporized portion of the sample and carries the sample vapors into the column.
- To achieve the basic separation with the sharpest peaks, the injector must **instantly vaporize** all the compounds of the sample and send the mixture onto the column.
- The injector block is heated to a temperature that is at **least 50 °C above** the sample component with the highest boiling point.

Types of Injection

On column injection

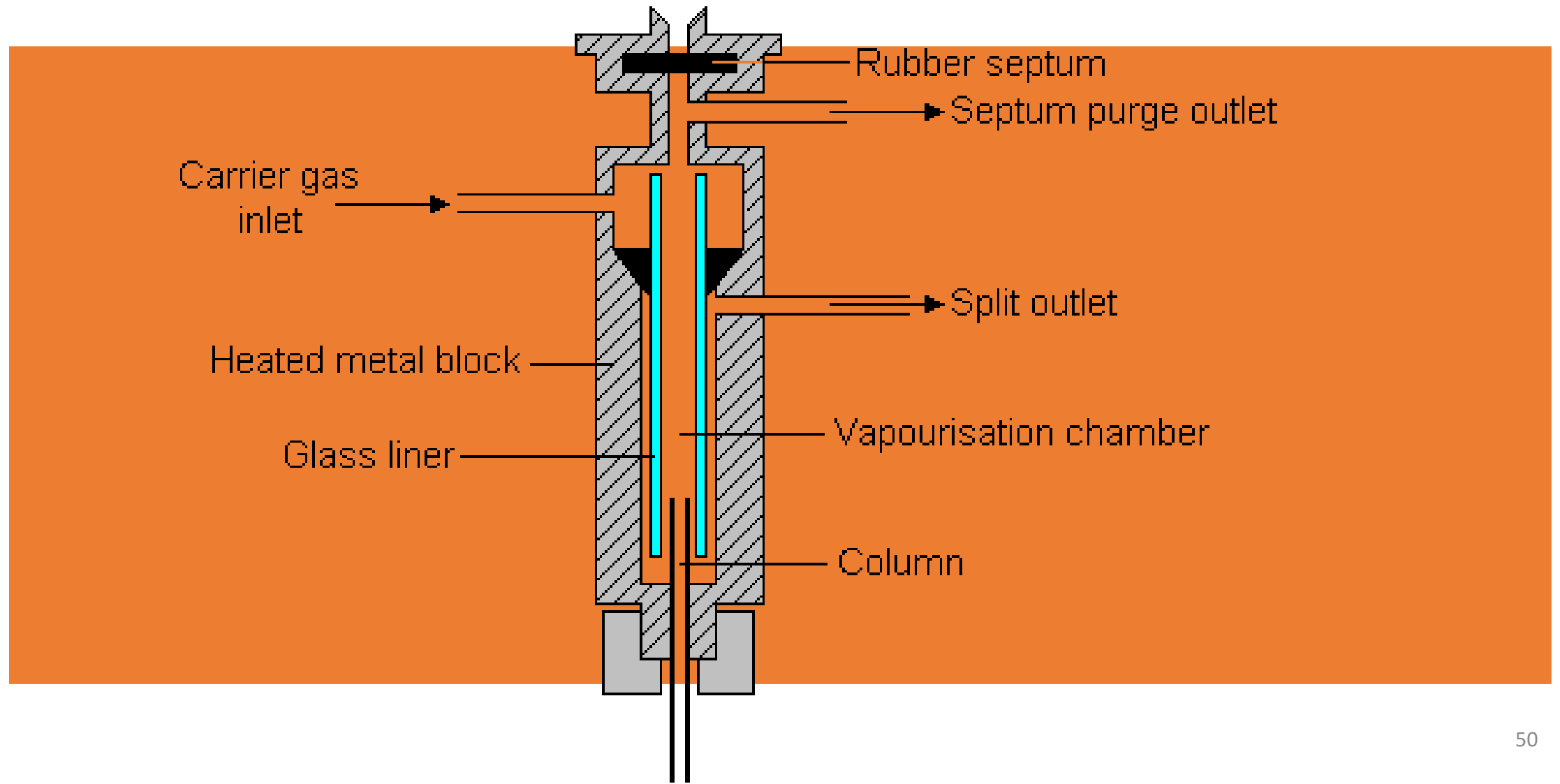
- A syringe with a very fine quartz needle is used. Air (Carrier Gas) cooled to -20°C below the bpt of the sample. After then the warmer air is circulated to vaporize the sample.
- used widely in packed-column GC, less in capillary GC
- **for thermally unstable compounds, high Mwt compounds, polar compounds, Trace level analysis, biological and natural samples (essential oils, food extracts, biological fluids)**
- sample - deposited directly on the column
 - Good for *thermally unstable compounds*
 - Good for *quantitative analysis at low concentrations*
 - entire sample reaches the detector
- For packed columns, liquid samples of **$0.1\mu\text{L}$ to $10\mu\text{L}$** are injected, while a gas samples of **1 to 10 ml** are injected.

- **Capillary columns** require the use of a special injector to avoid overloading the column with sample.
- Several capillary injectors are available, the most common of which is a **split/splitless injector**.

Split injection

- A technique for injecting samples onto a capillary column in which only a small portion of the sample enters the column.
- it splits the volume of sample stream into **two unequal flows** by means of a needle valve , and allow the **smaller flow to pass on to the columns** and the **bigger part is allowed to be vented to the atmosphere**.
- This technique is **not suitable when highest sensitivity** is required.
- Only about **0.1–1%** of the sample enters the column, with the remainder carried off as waste.
- **Not for analytes with a wide range of boiling points** (some may swept out the split vent before volatilization)

The split / splitless injector



Splitless injectors

- They allow **all of the sample to pass** through the column for loading.
- Sample should be very dilute to avoid overloading of the column and a high capacity column such as SCOT or heavily coated WCOT columns should be used.
 - Sample - vaporized in the injector and **ALL** of the sample is swept onto the column by the carrier gas
 - Sample spends a large amount of time in the injector
 - Best for trace (1 -100 ppm range) concentrations of high boiling point analytes
 - **extra time in the injector helps volatilize the analytes**



Automatic injectors

For improving the reproducibility and if a large number of samples are to be analyzed or operation is required without an attendant, automatic injectors are used.

- The solid samples are introduced as a solution or in a sealed glass ampoule, crushed in the gas stream with the help of a gas tight plunger, and the sample gets vaporized and flows into column under the influence of carrier gas.

Columns

- A chromatographic column provides a location for physically retaining the stationary phase.
- The column's construction influences
 - ✓ *the amount of sample that can be handled,*
 - ✓ *the efficiency of the separation,*
 - ✓ *the number of analytes that can be easily separated, and*
 - ✓ *the amount of time required for the separation.*

❑ There are two general types of columns:

1. Packed columns:- In **GLC**, they are densely packed with finely divided, inert, solid support material (diatomaceous earth) coated with liquid stationary phase.

GLC- supports like diatomaceous earth, chromosorb P and W are commonly used

- Stationary phase in GLC are either liquid or low melting solids.
- The liquid stationary phase should be ***non-volatile and thermally stable*** at the operation temperature of the column.
- Can be polar (e.g. polystyres, carbowaxes, amines etc) or non polar (hydrocarbons, silicon oils, and greases)

GSC, the columns are packed with adsorbents or porous polymers.

➤ Length- 1.5 - 10m

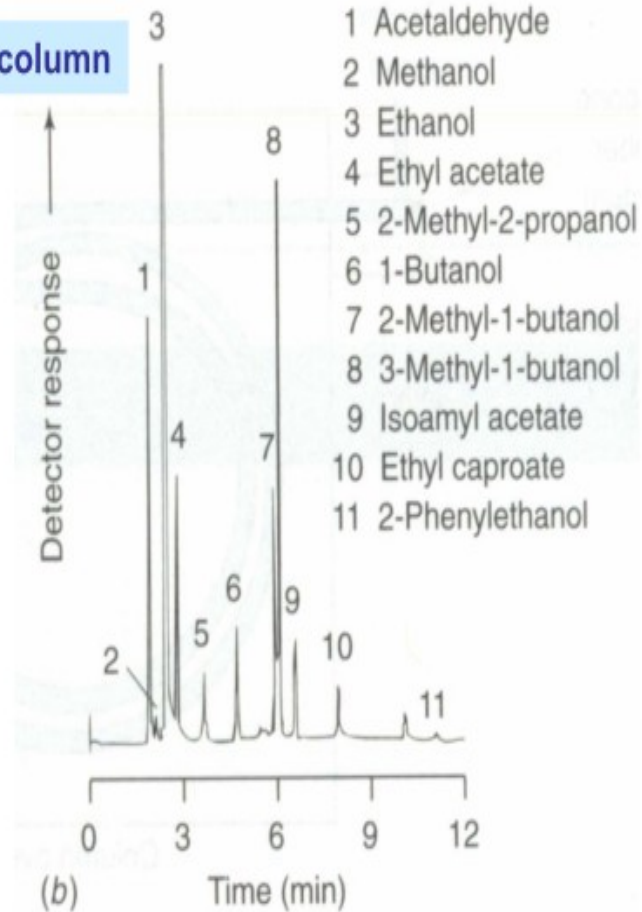
➤ internal diameter- 2 - 4mm.



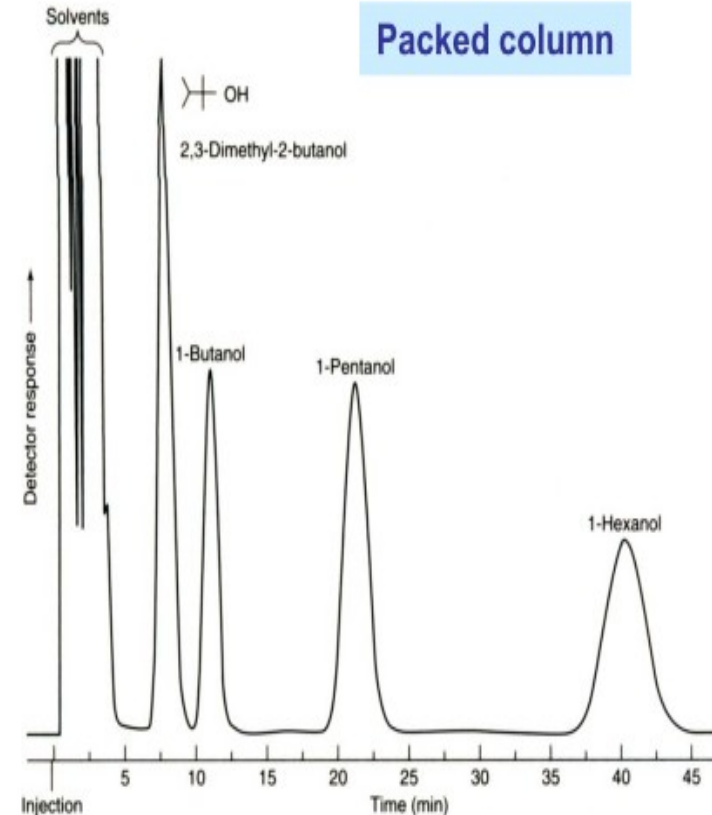
Packed columns...

- Greater sample capacity
- Broader peaks, longer retention times and less resolution
 - **Improve resolution by using small, uniform particle sizes**

Open tubular column

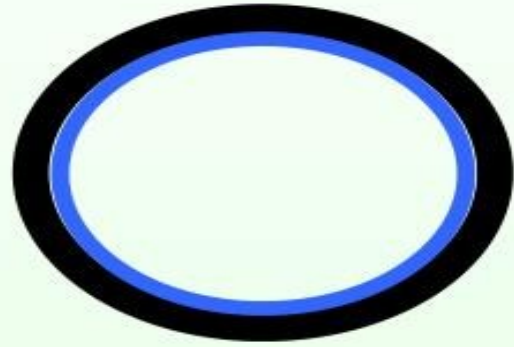


Packed column



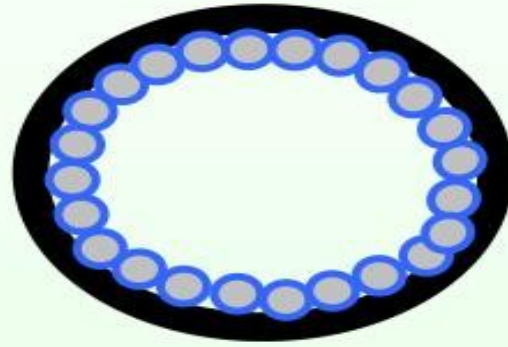
2. Capillary columns

- length ranges from 10-100m
- inner diameter is usually 0.1-0.5mm.
- ✓ **Wall-coated columns** - consist of a capillary tube whose walls are coated with liquid stationary phase.
- ✓ **Support-coated columns**- the inner wall of the capillary is lined with a thin layer of support material such as diatomaceous earth, onto which the stationary phase has been adsorbed.
- It is also known as PLOT (porous-layer open tubular columns).
- SCOT columns are generally less efficient than WCOT columns.
- Both types of capillary column are more efficient than packed columns.



wall-coated
open tubular column
(WCOT)

FSOT: fused-silica, OTC



support-coated
open tubular column
(SCOT)



porous-layer
open tubular column
(PLOT)

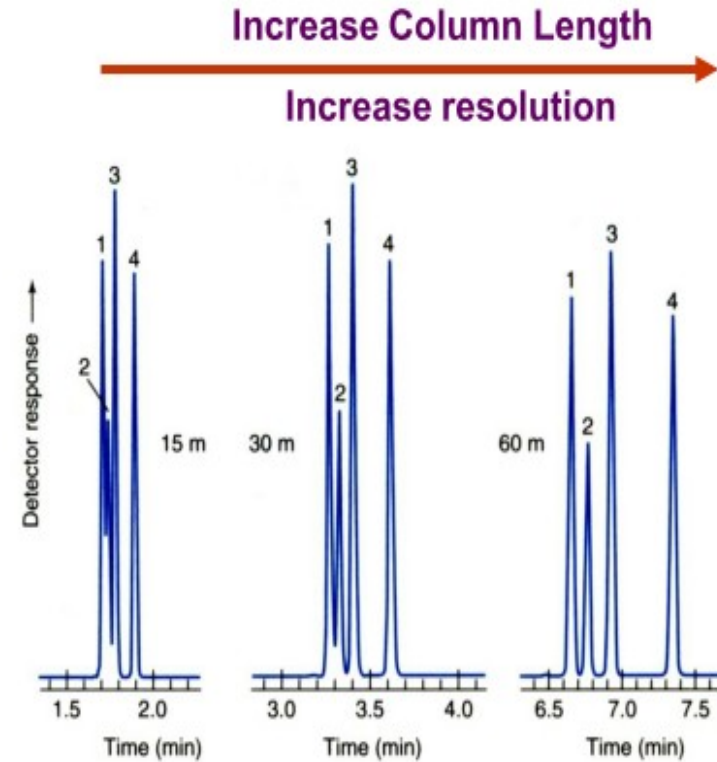
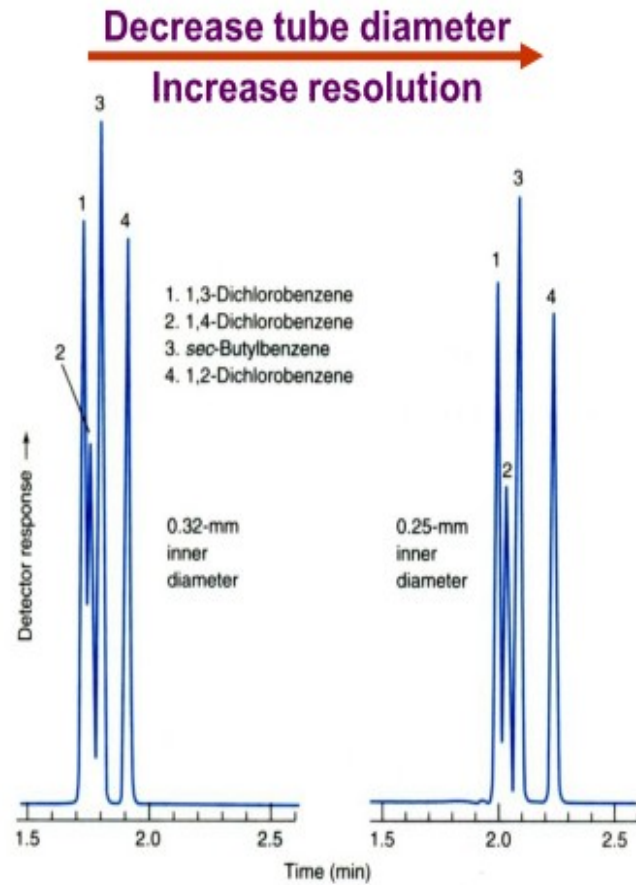
Decreasing thickness of stationary phase leads to

1. decreased plate height
2. decreased retention time
3. decreased capacity for analyte



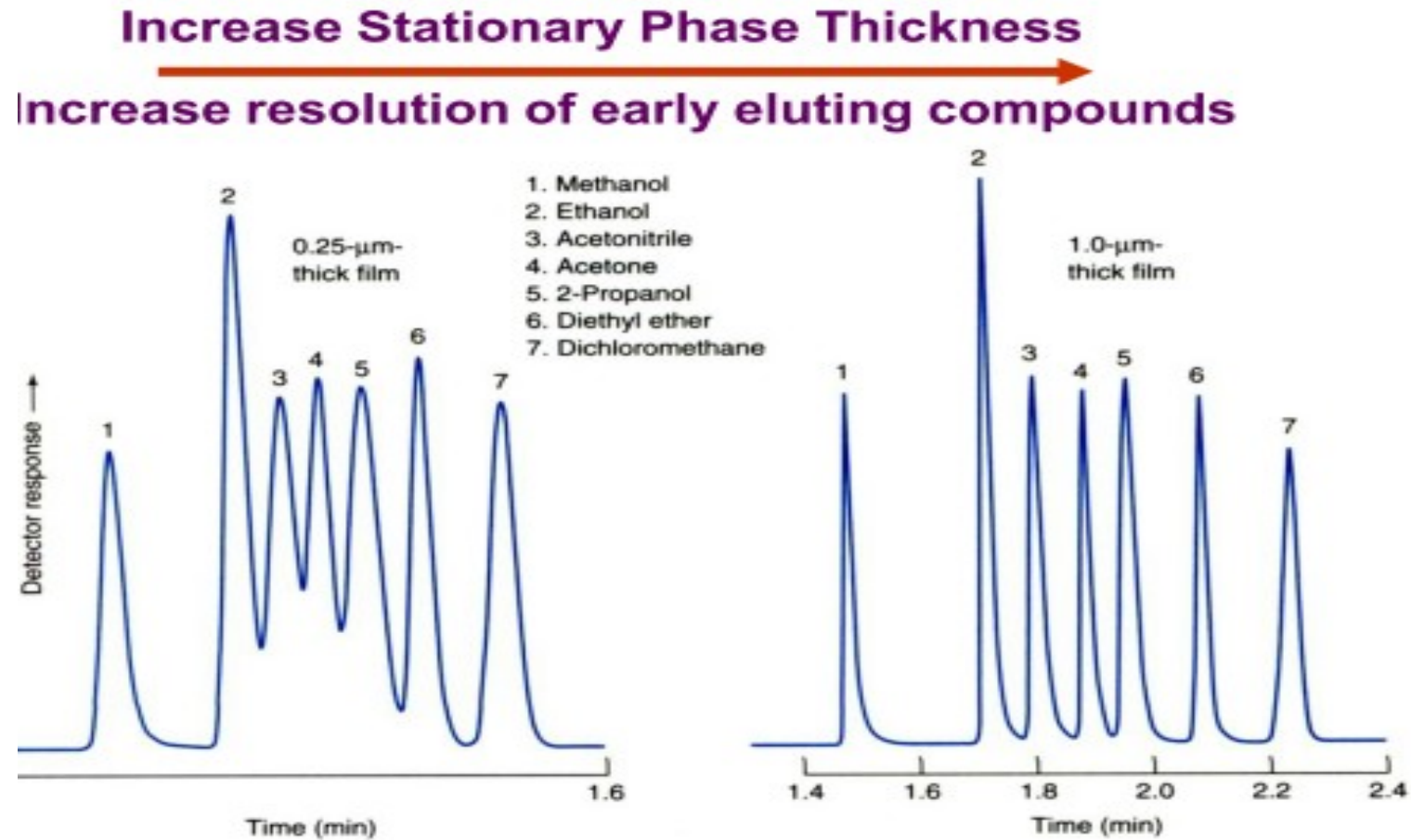
capillary column...

- Increasing Resolution



Capillary column...

- Increasing Resolution...



Difference between packed and capillary column

Packed Column	Capillary Column
Contains fully packed stationary phase made of fine particles	SP is coated on the inner surface
Has a packed SP	Coated SP
Require large amount of sample	Requires small amount of sample
Have high pressure inside the column	Less pressure inside the column
Short	Long
Dimeter can be several millimeters	1mm diameter
Low efficiency	High efficiency
Comparatively poor resolution	Higher resolution
Less expensive	More expensive
Better for separating non-polar samples Since there tube is stainless steel	Better for separating polar samples since there tube is glass

Support Material

It's main function is to provide mechanical support to the liquid phase.

- ❑ An ideal support should have: a **large surface area**, **chemically inert**, should get **uniformly wet** with liquid phase, should be **thermostable**.

Commonly used solid phases are: diatomaceous earth or kieselguhr, glass beads, porous polymers, sand, etc.

- ❑ ***Liquid phase:-***

It should have the following requirements:

- ✓ It should be non-volatile
- ✓ Should have high decomposition temperature
- ✓ Should be chemically inert
- ✓ Should possess low vapour pressure at column temperature
- ✓ Should be chemically and structurally similar to that of the solute i.e., polar for polar solute.

SP(GLC)

- Selectivity in gas chromatography is influenced by the choice of stationary phase
- **Elution order** in GLC is determined primarily by the solute's boiling point and, to a lesser degree, by the solute's interaction with the stationary phase.
- Solutes with *significantly different boiling points* are easily separated.
- On the other hand, two *solutes with similar boiling points* can be separated only if the stationary phase selectively interacts with one of the solutes.

- In general, non-polar solutes are more easily separated with a non-polar stationary phase, and polar solutes are easier to separate using a polar stationary phase.
- The main criteria for selecting a stationary phase are that it should be
 - **chemically inert,**
 - **thermally stable**
 - **of low volatility, and**
 - **of an appropriate polarity for the solutes being separated.**
- The following table shows some of the stationary phases used in GLC.

Stationary Phase	Common Trade Name	Maximum Temperature, °C	Common Applications
Polydimethyl siloxane	OV-1, SE-30	350	General-purpose nonpolar phase; hydrocarbons; polynuclear aromatics; drugs; steroids; PCBs
Poly(phenylmethyldimethyl) siloxane (10% phenyl)	OV-3, SE-52	350	Fatty acid methyl esters; alkaloids; drugs; halogenated compounds
Poly(phenylmethyl) siloxane (50% phenyl)	OV-17	250	Drugs; steroids; pesticides; glycols
Poly(trifluoropropyldimethyl) siloxane	OV-210	200	Chlorinated aromatics; nitroaromatics; alkyl-substituted benzenes
Polyethylene glycol	Carbowax 20M	250	Free acids; alcohols; ethers; essential oils; glycols
Poly(dicyanoallyldimethyl) siloxane	OV-275	240	Polyunsaturated fatty acids; rosin acids; free acids; alcohols

➤ *Equilibrium of the column:-*

- The packed columns are equilibrated before introduction of the sample. This is done by allowing continuous flow of heated carrier gas through the columns for a specific duration of time (24hrs) at prescribed temperature.
- Ideally prepared and **conditioned columns** show a zero base line on the recorder upon passage of carrier gas alone.

➤ *Column temperature:-*

- Control of the column's temperature is critical to attaining a good separation in gas chromatography
- For this reason the column is located inside an oven.
- This can be controlled by jackets equipped with vapors of a boiling liquid, electrically heated metal blocks or circulating air baths.

In an isothermal separation the column is maintained at a **constant temperature**, the choice of which is dictated by the solutes

- ❖ One difficulty with an isothermal separation is that **a temperature favoring the separation of low-boiling solutes may cause unacceptably long retention times for higher boiling solutes.**
- ✓ Compounds of low B.Pt- eluted at lower temperature
- ✓ Compounds of high B.Pt- boils at higher temperature resulting in broader and shallower peaks, require temperature programming.

Gas Chromatography...

Column temperature...

- **Temperature programming**
 - Ovens capable of temperature programming provide a solution to this problem.
 - The initial temperature is set below that for the lowest boiling solute.
 - As the separation progresses, the temperature is slowly increased at either a uniform rate or in a series of steps.

Gas Chromatography...

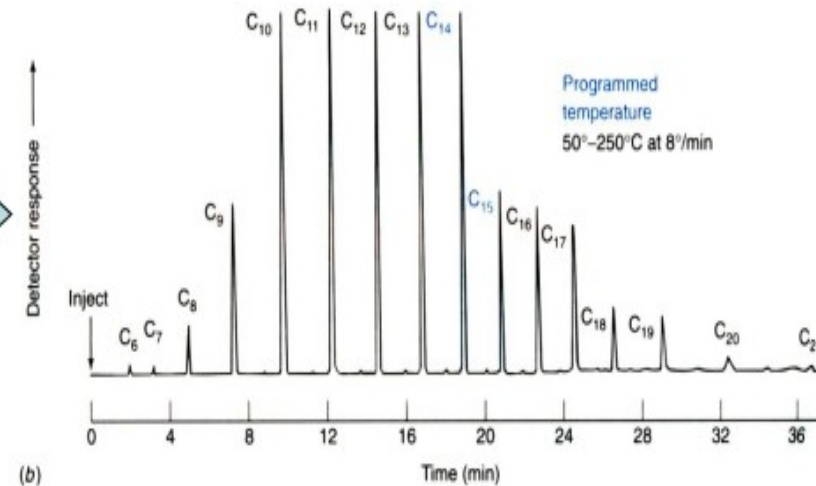
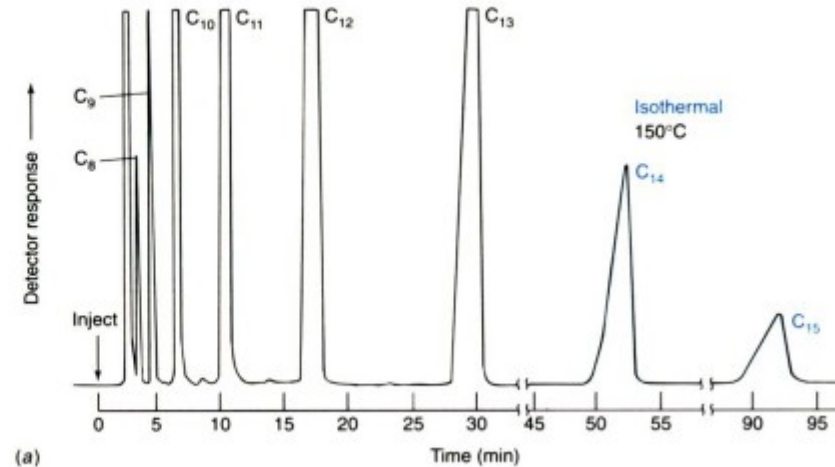
Column temperature...

Improving Column Efficiency

➤ Temperature programming:

- **Temperature is raised during the separation (gradient)**
- **increases solute vapor pressure and *decrease* retention time**

Temperature gradient improves resolution while also decreasing retention time



Detectors

- As each compound exits the column, it enters the detector.
- The detector interacts with the compounds based on some physical or chemical property.
- Some detectors *respond to every compound* while others *respond only to a selected group* of compounds.
- The interaction generates an electrical signal **whose size corresponds to the amount of the compound.**
- The detector signal is then sent to a recording device for plotting.

Detectors...

- The ideal detector has several desirable features, including:
 - ✓ **low detection limits**
 - ✓ **A linear response over a wide range of solute concentrations**
 - ✓ **Responsiveness to all solutes or selectivity for a specific class of solutes**
 - ✓ **An insensitivity to changes in flow rate or temperature.**

Gas Chromatography...

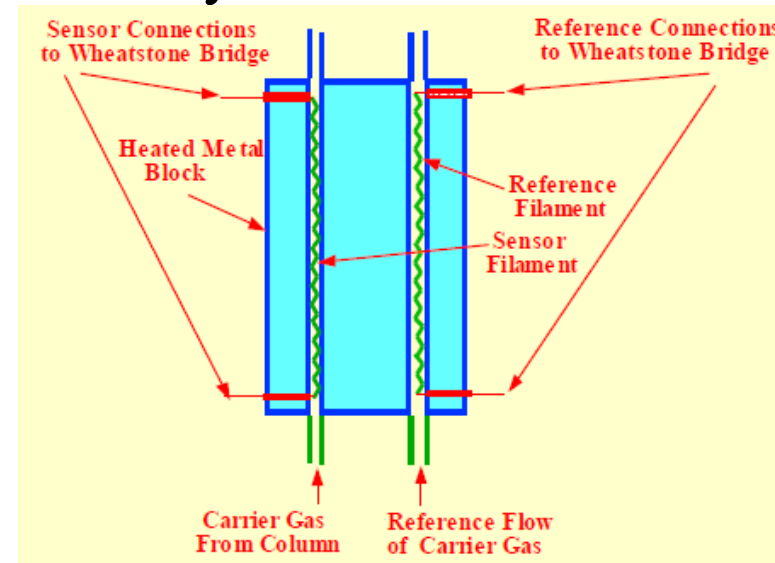
There are different types of detectors which can be used in GC. These include

- Thermal conductivity detector
- Flame ionization detector
- Electron capture detector
- Nitrogen-phosphorus
- Mass spectroscopy
- Infrared spectroscopy

Gas Chromatography...

Thermal conductivity detector

- Compares the temperature of two heated filaments: a **sample filament** and a **reference filament**
- The filaments are made from an alloy whose resistance varies with temperature. The filaments are heated by electric current.



TCD...

- Initially, carrier gas (**He**) flows through both the reference and sample filament.
- When a sample molecule with lower thermal conductivity than the carrier gas exits the column and flows across the sample filament, the thermal conductivity decreases.
- This decrease in conductivity **unbalances the Wheatstone bridge** and generates a peak as sample molecules transit through the detector.

TCD...

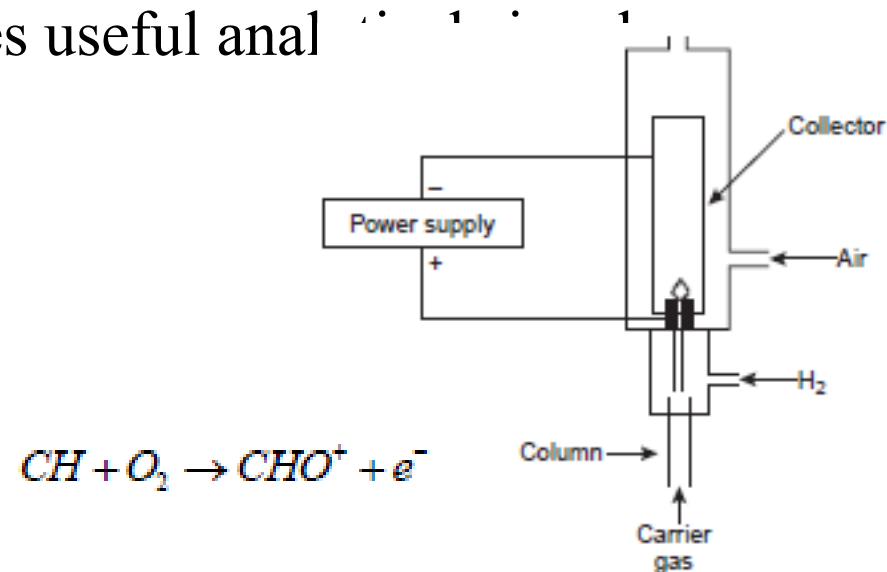
A TCD detector has the advantage of **universality**, since it gives a signal for any solute whose thermal conductivity differs from that of helium.

- Another advantage is that it gives a **wide linear response** for solute concentrations
- The detector also is **nondestructive**, making it possible to isolate solutes with a post detector cold trap.
- **It is simple and low cost.**
- Thermal conductivity detector's **detection limit is poor** in comparison with other popular detectors.
- **TCD filaments quickly burn out** if they are heated in the absence of carrier gas.

Gas Chromatography...

B. Flame ionization detector

- Burns the effluent coming off the column in hydrogen-air flame.
- The combustion of an organic compounds or hydrocarbons (C-H) produces ions.
- The ions are collected **on a cathode** and produce current
- When amplified, this current provides useful anal



Flame ionization detector...

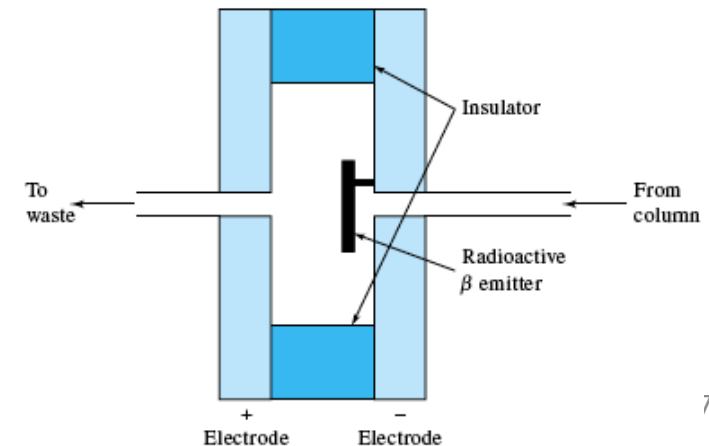
- Most carbon atoms, except those in carbonyl and carboxylic groups, generate a signal, making the FID an almost universal detector for **organic compounds**.
- **WILL NOT detect non-C containing compounds**
- Requires H₂ supply (tank or generator) and O₂ supply (compressed air)
- H₂ carrier gas can be used, eliminating the need for a supply for the detector.
- Most inorganic compounds and many gases, such as H₂O and CO₂, cannot be detected, making the FID detector ideal for the analysis of atmospheric and aqueous environmental samples.

Flame ionization detector...

- Advantages of the FID include a detection limit that is approximately two to three orders of magnitude smaller than that for a thermal conductivity detector.
- Wider linear range than TCD
- **The sample, of course, is destroyed when using a flame ionization detector.**

C. Electron capture detector

- The **electron capture detector** is an example of a selective detector.
- The detector consists of a beta emitter (a beta particle is an electron) such as ^{63}Ni .
- The emitted electrons **ionize the mobile phase**, which is usually N_2 , resulting in the production of additional electrons that *give rise to an electric current between a pair of electrodes*



Electron capture detector...

- $\text{N}_2 + \text{e}^- \longrightarrow \text{N}_2^+ + \text{e}^- + \text{e}^*$
- 20 thermal e-s
- When a solute with a high cross section for the capture of electrons (electronegative) elutes from the column, the electric current decreases.

Gas Chromatography...

C. Electron capture detector...

- This decrease in electric current serves as the signal.
- The ECD is highly selective toward solutes with *electronegative functional groups, such as halogens, and nitro groups* and is relatively insensitive to amines, alcohols, and hydrocarbons.
- Widely used for the determination of pesticides, herbicides and PCBs in environmental samples.
- Although its detection limit is excellent, its linear range is small.
It is Non-destructive

Gas Chromatography...

5. Data systems

- In the early days of GC, **strip chart recorders**, were the mainstay of collecting chromatograms.
- Today, digital revolution has come up with *electronic recording-integrators or microcomputers*.
- Signal from the detector amplifier is digitized and stored to disk allowing enormous convenience in retrieving and replaying results.
- *This means that peak retention times, peak areas, etc. are automatically reported.*

Gas Chromatography...

□ Applications of GC

- Gas chromatography is widely used for the analysis of a diverse array of samples in environmental, clinical, pharmaceutical, biochemical, forensic, food science, and petrochemical laboratories.
- Can be classified into
 - Quantitative application
 - Qualitative application

Gas Chromatography...

A. Qualitative Application

- Gas chromatography can be used for qualitative purposes.
- When using an IR or a mass spectrometer as the detector, the available **spectral information** often can be used to identify individual solutes.
- With conventional non spectroscopic detectors, other methods must be used to identify the solutes.
- One approach is **to spike the sample** by adding an aliquot of a suspected analyte and **looking for an increase in peak height/area/**.
- **Retention times** also can be compared with values measured for standards, provided that the operating conditions are identical.

Gas chromatography...

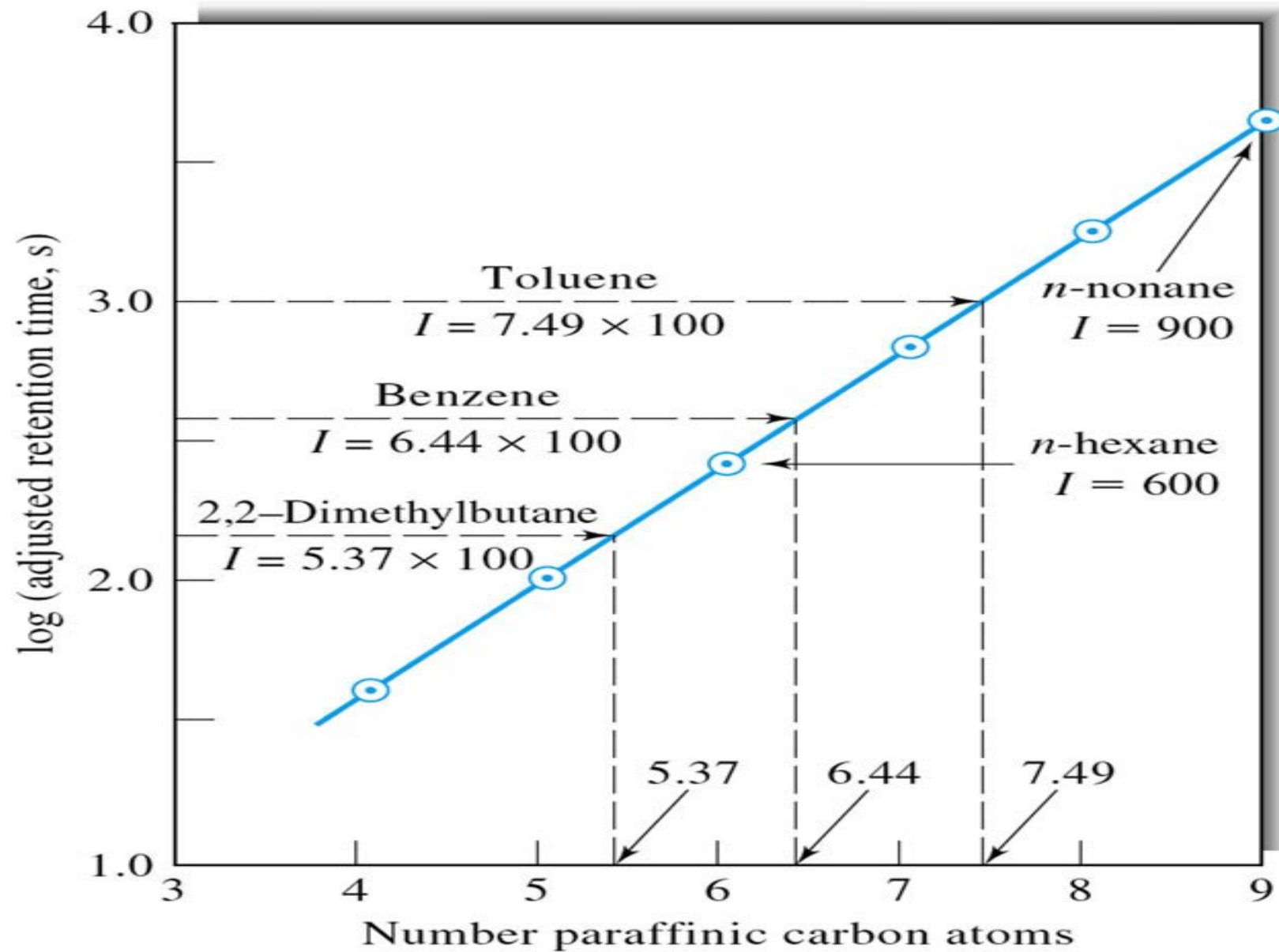
A. Qualitative Applications...

- Because of the difficulty of exactly matching such conditions, tables of **retention times are of limited utility.**

Gas chromatography...

Kovat's retention index (I)

- I for an analyte is a measure of the rate at which it is carried through a column compared with the rate of movement of two normal alkanes one that moves faster than the analyte and the other that moves more slowly.
 - Kovat defined the **retention index, I** , for a normal alkane as *100 times the number of carbon atoms*; thus, the retention index is 400 for butane and 500 for pentane.
 - To determine the retention index for another compound, its adjusted retention time is measured relative to that for the normal alkanes eluting just before and after.



A. Qualitative Applications...

For example, a compound eluting between butane and pentane has a retention index between 400 and 500.

- The exact value for the compound's retention index, *I_{cpd}*, is given as

$$I = 100 \left[n + (N - n) \frac{\log \text{tr}'(\text{Unknown}) - \log \text{tr}'(n)}{\log \text{tr}'(N) - \log \text{tr}'(n)} \right]$$

- *I* = Kovat's retention index
- *n* = the number of carbon atoms in the smaller n-alkane.
- *N* = the number of carbon atoms in the larger n-alkane
- *tr* = the retention time
- Provides means to compare the performance of different columns

B. Quantitative Application

- In a quantitative analysis, the **height or area** of an analyte's **chromatographic peak** is used to determine **its concentration**.
- Calibration curves are usually constructed by analyzing a series of external standards and *plotting the detector's signal as a function of their known concentrations*.
- As long as the injection volume is identical for every standard and sample, calibration curves prepared in this fashion give both accurate and precise results.

HIGH PERFORMANCE LIQUID CHROMATOGRAPHY

- **HPLC** is basically a highly improved form of column chromatography.
 - It is a sophisticated form of liquid chromatography.
 - Fine column packing particles are used to increase the efficiency.
 - The decrease in particle size of the packing materials makes the use of pumping system necessary.
 - It is often necessary to use samples less than 20 microgram.
 - Sensitive detectors and data handling systems required to analyze the column effluent.
 - Improved detection methods.

- There are two variants in HPLC depending on the relative polarity of the solvent and the stationary phase.

1. Normal phase HPLC- uses polar stationary phase and non-polar mobile phase.

✓ Polar compounds in the mixture being passed through the column will stick longer to the polar silica than non-polar compounds will.

✓ The non-polar ones will therefore pass more quickly through the column.

2. Reversed phase HPLC- the column is filled with silica which is modified to make it non-polar by attaching long hydrocarbon chains to its surface - typically with either 8 or 18 carbon atoms in them.

- Uses non-polar stationary phase and polar mobile phase
- There will be a strong attraction between the polar solvent and polar molecules in the mixture being passed through the column.
- A polar solvent is used - for example, a mixture of water and an alcohol such as methanol.

Reversed phase...

- There won't be as much attraction between the hydrocarbon chains attached to the silica (the stationary phase) and the polar molecules in the solution.
- Polar molecules in the mixture will therefore spend most of their time moving with the solvent.
- Non-polar compounds in the mixture will tend to form attractions with the hydrocarbon groups because of **van der Waals dispersion forces**.

Reversed phase...

- They will also be less soluble in the solvent because of the need to break hydrogen bonds as they squeeze in between the water or methanol molecules, for example.
- They therefore spend less time in solution in the solvent and this will slow them down on their way through the column.
- That means that now it is the polar molecules that will travel through the column more quickly.
- Reversed phase HPLC is the most commonly used form of HPLC.

Advantages of HPLC over GC

- No need for derivatization.
- Broader applicability
 - ✓ Polar
 - ✓ Non-Volatile
 - ✓ Thermally unstable
 - ✓ Better for large molecules
 - ✓ Macromolecules of biomedical interest
 - ✓ Labile Natural products
 - ✓ A wide variety of high molecular weight compounds like Proteins, nucleic acids, polysaccharides, synthetic polymers etc.

➤ Very difficult separations are often achieved by LC:

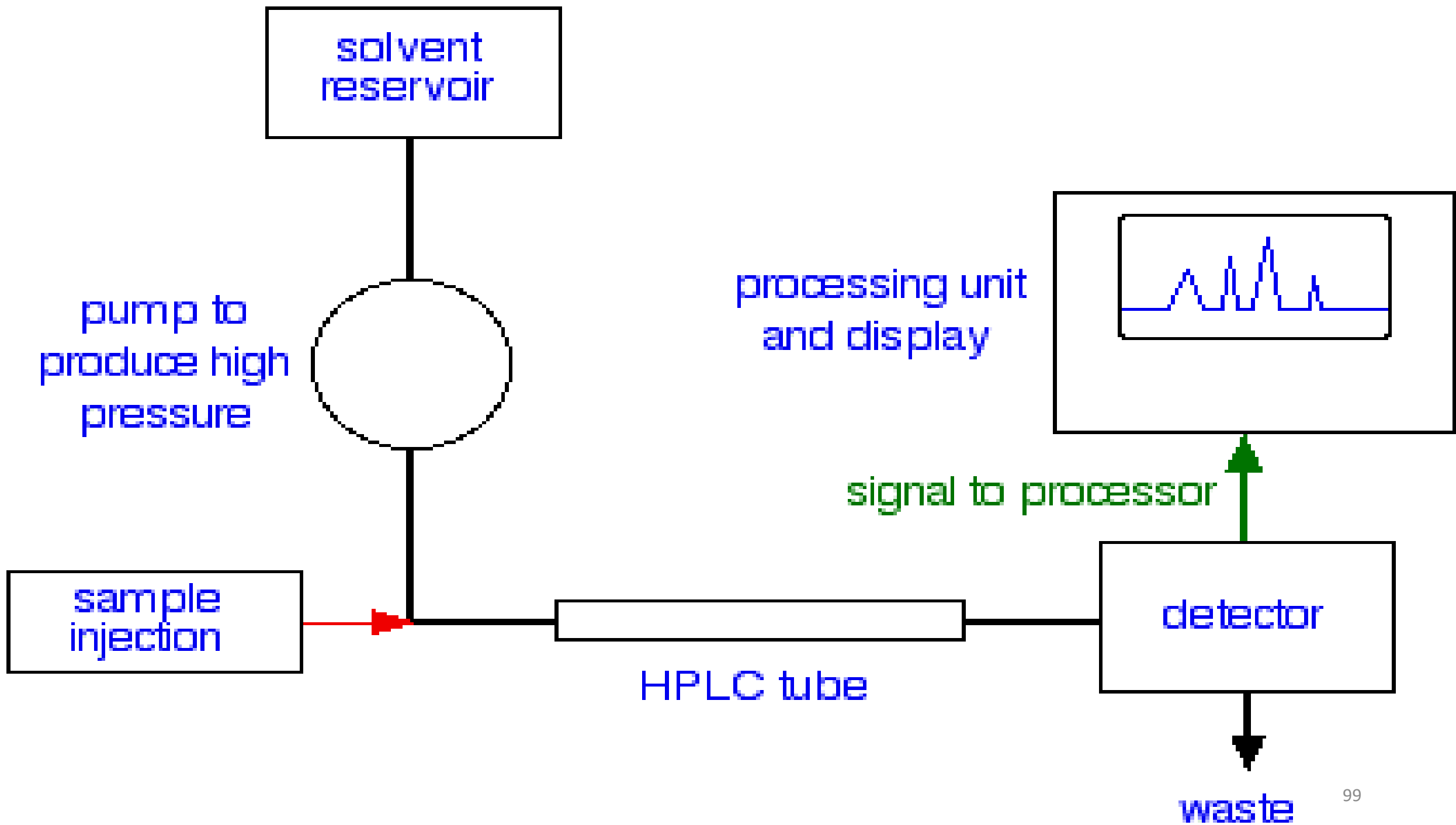
➤ **Reasons include**

- Liquid chromatographic separations are the result of interaction b/n sample molecules, stationary and mobile phase.
 - Interaction with the mobile phase is absent in GC.
 - Presence of many variables assist in controlling and improving separation.
- Greater variety of stationary phases are available for HPLC which in turn gives chance for wider variation of selective interactions there by resulting greater possibilities for separation.
- Chromatographic separation is generally enhanced at lowered temperature.
 - Low temperature makes intermolecular interactions more effective.
 - This favors HPLC that is usually performed at room temperature

- Relative ease of isolate recovery
 - Separated fractions can easily be collected in HPLC.
 - Recovery of separated components in GC is also possible but is generally less convenient and not quantitative.

Principles of HPLC

- Mobile phase is filtered, degassed and pumped through a chromatographic column.
- The sample is injected and loaded onto the column where the components of the mixture are separated.
- The effluent is monitored using a detector and recorded as peaks.



HPLC...

- A modern HPLC consists of
 - A solvent unit with degassing system and gradient mixers
 - High pressure pump and pressure gauge that can provide constant flow.
 - Injector/Injection port/
 - Column
 - High sensitivity detectors
 - High speed data acquisition and recorder systems
 - Low dispersion connecting tubes-connects valve to column and column to detector

HPLC...

1. Mobile phase reservoirs

- Designed to allow mobile phase degassing in situ.
- To eliminate dissolved gases particularly **oxygen**.
- Which might interact with the mobile phase or stationary phase in the column or
- May form a bubble in the detector thereby resulting in signal distortion.
- Degassing is accomplished in several ways, but the most common are
 - ✓ The use of a vacuum pump
 - ✓ Sparging with an inert gas, such as He, which has a low solubility in the mobile phase.
A helium blanket on the surface of the solvent prevents oxygen from dissolving in the solvent after degassing.

HPLC...

1. Mobile phase reservoirs...

- Particulate material capable of **clogging** the HPLC tubing or column is removed by filtering.
- If the instrument is not designed to do so, degassing and filtering can be completed before the solvents are placed in their reservoirs.
- *There are two basic types of mobile phase delivery*
 - *Isocratic elution*-When a separation uses a single mobile phase of fixed composition.
 - It is often difficult, however, to find a single mobile-phase composition that is suitable for all solutes.
 - Mobile phase that is suitable for early eluting solutes may lead to unacceptably long retention times for later eluting solutes

HPLC...

Gradient elution-solvent mixture whose composition is varied with time is used for elution.

- **Mixers**- provides very precise control of solvent composition.
 - The solvents might be first pumped individually and then mixed or (**High-pressure mixing system**)
 - They might be first mixed with the required proportion and then pumped (**Low pressure mixing system**)
- **Mobile phases**- the polarity of mobile phase might help in optimizing the retention time of the solutes and hence separation efficiency.
 - **Eg. In RP-HPLC**- retention time of non-polar compounds can be made to increase by increasing the polarity of mobile phase.

HPLC...

- **Choosing a Mobile Phase-** Several indices have been developed to assist in selecting a mobile phase, the most useful of which is the **polarity index**.
- The following table shows values for the polarity index, P' , of several commonly used mobile phases, in which larger values of P' correspond to more polar solvents.

Mobile Phase	Polarity Index (P')	UV Cutoff (nm)
cyclohexane	0.04	210
n-hexane	0.1	210
carbon tetrachloride	1.6	265
i-propyl ether	2.4	220
toluene	2.4	286
diethyl ether	2.8	218
tetrahydrofuran	4.0	220
ethanol	4.3	210
ethyl acetate	4.4	255
dioxane	4.8	215
methanol	5.1	210
acetonitrile	5.8	190
water	10.2	—

HPLC...

2. Pumps

The HPLC pump, often called a **solvent delivery system**, provides a precise flow of mobile phases of a specified composition to the column.

- In **HPLC**-long narrow columns packed with very fine particles are used.
- This packing makes it resistant to the flow of mobile phase
- High pressure pumps are needed to force solvents through packed stationary phase beds.

HPLC...

2.Pumps...

➤ The pressure required depends **on:-**

✓ **flow rate**

✓ **viscosity of the solvent**

✓ **size of the particles that form the stationary phase.**

➤ Devices for monitoring column input pressure are inserted in line b/n the pump and the column.

➤ **Pressure gauge** indicates if there has been plugging of the column (**increase in pressure**) or failure of the pumping system (**decrease in pressure**)

HPLC...

2. Pumps...

- Pumps should have components inert to the wide variety of solvents used in HPLC.
- Should have stable flow rate (consistent flow rate.)
- Should be able to provide accurate flow rates from 0.1 to 10 mL/min.
- Should have accurate control of flow rate changes (0.5% or less relative error).
- Should contain corrosion- and leak-resistant components.

HPLC...

2.Pumps...

- The pump directly affects the retention time, reproducibility and detector sensitivity.
- The pump should be capable of delivering the mobile phase at flow rates of 1–5 ml/min and pressure of 5000 psi.

There are mainly two types of pumps.

- ✓ Constant pressure pumps
Example: Displacement type pumps.
- ✓ Constant volume pumps
Example: Reciprocating pumps and syringe pumps.

HPLC...

2.Pumps...

- ✓ **Displacement type pumps**
- ✓ **The reciprocating type pumps**
- ✓ **A pressure vessel pumps (pneumatic pumps)**
- ✓ **Syringe type pumps**

Among these reciprocating pumps are widely used.

Pumps...

Pump Type	Flow Type	Reservoir Capacity	Advantages	Limitations
Displacement Pump	Non-pulsating	Limited	Smooth flow, avoids fluctuations	Short runs, limited reservoir
Reciprocating Pump	Pulsating	Unlimited	High pressure, continuous operation, widely used	Requires pulse dampeners to minimize fluctuations
Pressure Vessel Pump	Non-pulsating	Limited	Stable flow, pneumatic operation	Short runs, reservoir limits
Syringe Pump	Non-pulsating	Limited	Precise, reliable, highly accurate	Expensive, limited solvent capacity

Pumps...

- ❖ Reciprocating Pumps are the most common.
- ✓ Unlimited Reservoir – Can supply continuous mobile phase for extended runs.
- ✓ High Pressure Handling – Essential for HPLC, where pressures can exceed 4000 psi.
- ✓ Flow Rate Control – Can adjust solvent flow to match different analytical methods.
- ✓ Reliable & Affordable – Compared to syringe pumps, reciprocating pumps offer high precision at a reasonable cost

HPLC...

2.Pumps...

A. Displacement type pumps: It consists of screw or plunger is driven by the motor.

- The rotatory movement of the pump provides the continuous movement of the mobile phase.
- This is propelled by the revolving screw and pushes solvent through the needle into the column.
- The advantage of this pump is the independent of the back pressure and simple operation.

HPLC...

2.Pumps...

B. Pneumatic pumps

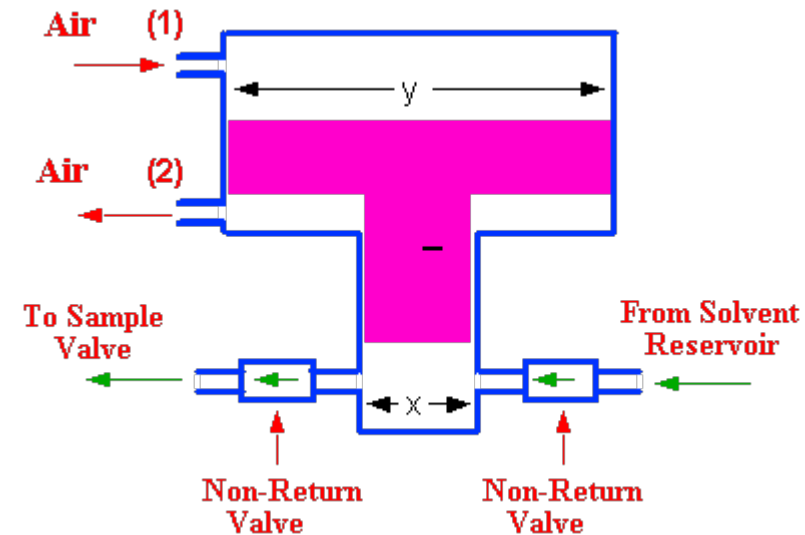
- Pneumatic pumps are the simplest where the mobile phase is pushed out of the mobile-phase container by the pressure from the gas cylinder by a piston.
- The flow is dependent on the back pressure of the column.
- Usually the flow is limited to pressures below 2000 psi.

HPLC...

2. Pumps...

B. Pneumatic pumps

- The pump operates in the following manner.
- Air enters port (1) and applies a pressure to the upper piston that is directly transmitted to the lower piston.
- liquid will flow out of the left hand side non-return valve as shown in the diagram
- At the extreme of movement a micro switch is activated and the air pressure transferred from port (1) to port (2).



HPLC...

2. Pumps...

Pneumatic pumps...

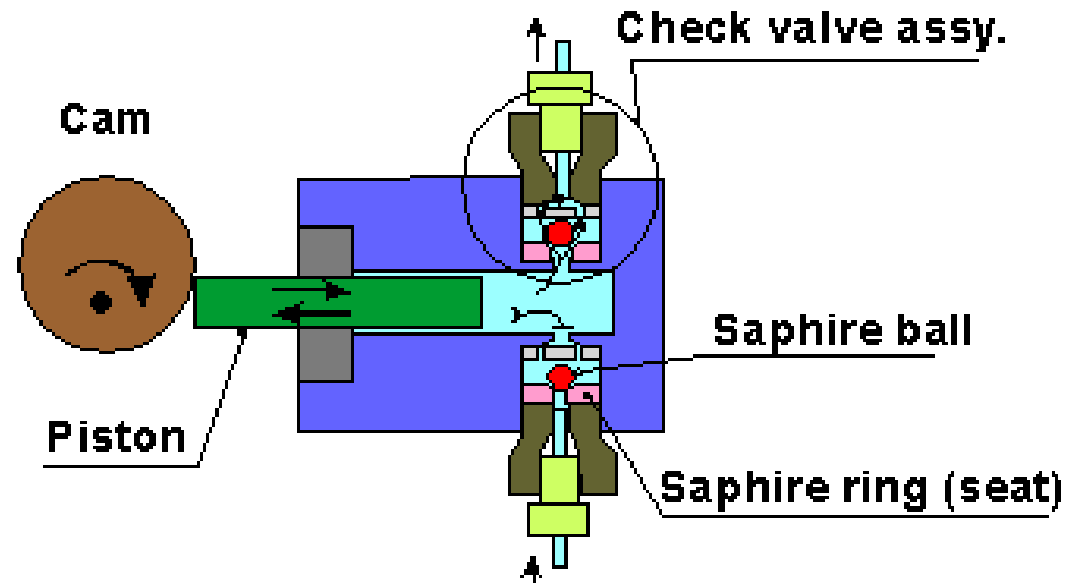
- The piston now moves in the opposite direction and draws mobile phase through **the right hand** non-return valve refilling the cylinder with solvent.
- Again on reaching the **limit of the piston movement** a second micro switches the air supply from port (2) back to port (1) and the process repeated.
- **Advantages**-low cost, simple operation, pulse free flow and generates high pressure.
- **Limitation-flow rate is not constant**
 - Is used in column packing due to the unnecessary high pressure
 - limited volume capacity such as 70 ml

HPLC...

2. Pumps

Reciprocating pumps

- With this pump there is no limit on the reservoir size or operating time as commonly found with other pumps.
- This type of pump is electrically driven by a motor, which moves back and forth within hydraulic chamber.



HPLC...

2. Pumps

Reciprocating pumps

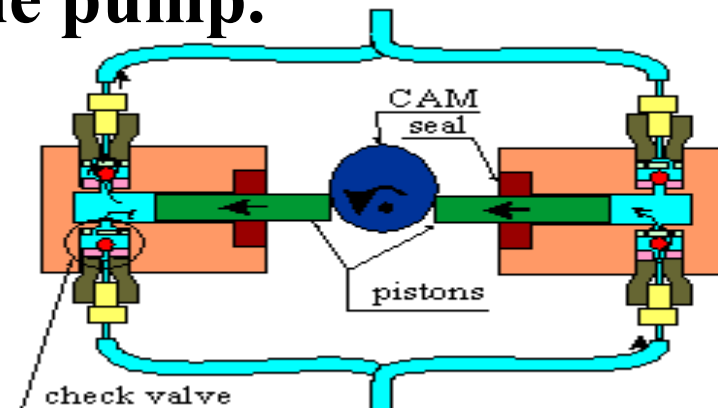
- CAM pushes piston back and force.
- **When the piston is moving backwards** it sucks the eluent through the inlet check valve (on the bottom).
- The sapphire ball is lifted and opens the path for the eluent.
- **When the piston moves forward**, the liquid pushes the inlet ball down and closes the path, but the outlet ball is lifted and opens the outlet valve (upper).
- The main disadvantage of this type of pump is sinusoidal pressure pulsations which lead to the necessity of using **pulse dampers**.

HPLC...

2. Pumps...

Reciprocating pumps

- A more efficient way to provide a constant and almost pulse free flow is the use of **dual-headed reciprocating pumps**.
- Both pump chambers are driven by the same motor through a common eccentric cam; this common drive allows one piston to pump while the other is refilling.
- As a result, the two flow-profiles overlap each other significantly **reducing the pulsation downstream of the pump.**
- But still require pulse dampers



HPLC...

Syringe Pumps

- Many early HPLC pumps in the 1960s and 1970s were syringe pumps (such as the, Varian 8500 Dual Syringe Pump System in 1975).
- The syringe pump is ideally suited for a perfectly constant flow without pulsation against a constant load.
- The gradient can be formed by using two or three syringe pumps.

HPLC...

2.Pumps...

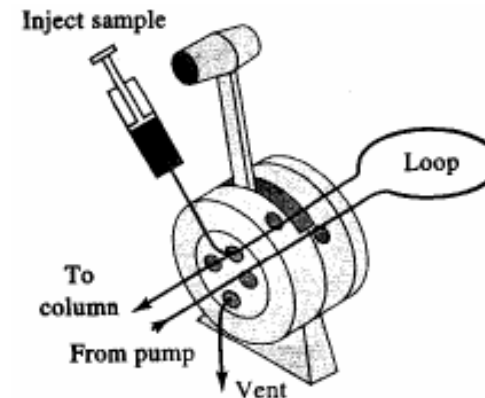
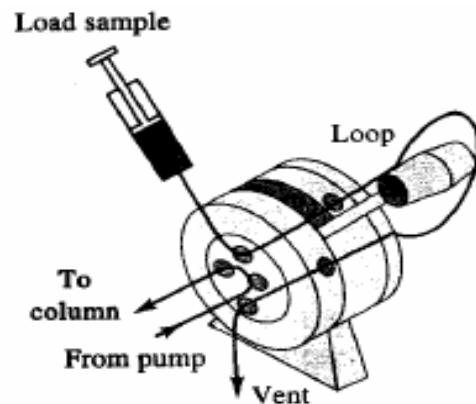
The big disadvantages are:

- the cumbersome screw-driven syringe mechanisms (fill volumes of 250 –1000 mL) needed to provide common analytical flow rates for each chromatographic run,.
- slow syringe retraction process before each sample injection
- ❖ Currently, syringe pumps are rarely used
 - ✓ except for micro or nano HPLC instruments (Eksigent nanoLC 400), and portable units (Axcend Focus LC)
 - ✓ the smaller flow rates allow for more compact pump designs, or post-column derivatization.

HPLC...

3. Sample injection

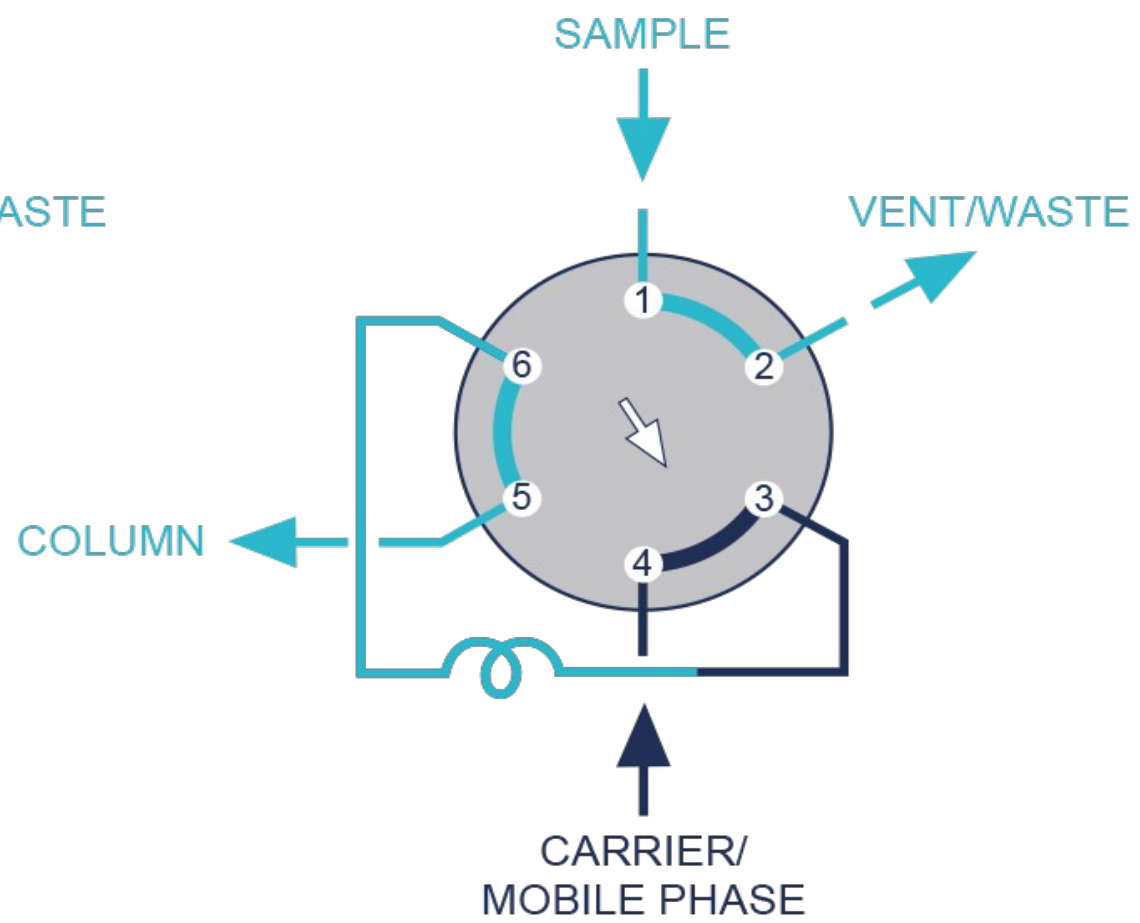
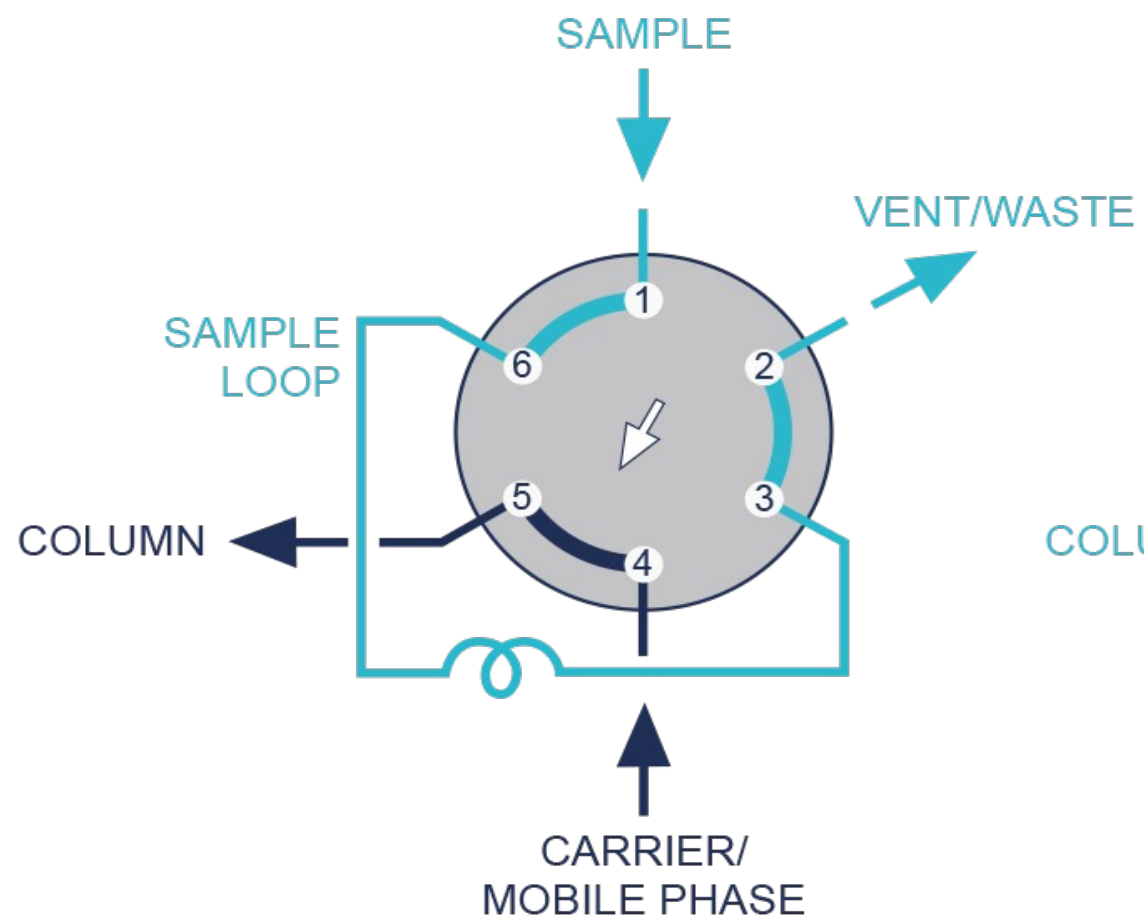
- The typical operating pressure of an HPLC is sufficiently high that it is impossible to inject the sample in the same manner as in gas chromatography.
- Instead, the sample is introduced using a **loop injector**
- Sampling loops are interchangeable, and available with volumes ranging from **0.5 μL to 2 mL**.
- **Two positions**-load and injection positions



HPLC...

3. Sample injection...

- **In the load position the sampling loop is isolated from the mobile phase and is open to the atmosphere.**
- A syringe with a capacity several times that of the sampling loop is used to place the sample in the loop.
- Any extra sample beyond that needed to fill the sample loop exits through the waste line.
- After loading the sample, the injector is turned to **the inject position**. In this position the mobile phase is directed through the sampling loop, and the sample is swept onto the column.
- The sample loop must allow delivery of very constant sample size. There should not be dispersion, adsorption and desorption.



HPLC...

4. HPLC Columns

- Columns are made up of stainless steel or heavy glass which with stands to the high pressure.
- Columns are usually narrow tubes packed with the 25 μm particles. The internal portion of the column should be smooth and uniform.
- Column dimensions in the range from 10 to 25 cm long and about 4.6 mm internal diameter.

The columns are of three types:

- ✓ Analytical columns.
- ✓ Preparative columns.
- ✓ Guard column

HPLC...

Column...

- **The guard column** is placed before the analytical column, protecting it from contamination.
 - Placed b/n the injector and the analytical column
 - Are shorter in length (3cm)
 - Contain the same stationary phase as the analytical column
 - Frequently replaced when required
 - Remove particulate matters from the solvent there by elongating the life of the analytical column

HPLC...

4. HPLC Columns...

Stationary phases

- Pellicular or porous packing materials are usually used. Pellicular packings are nonporous glass or polymer beads ranging from 30 to 40 μm .
- Porous packings are mostly silica based with particle diameters from 3 to 10 μm .
- ✓ Based on shape-could be **spherical or irregular**
- ✓ Based on porosity- **porous and superficially porous**
- ✓ Spherical column packing produces homogeneous stationary phase there by minimizing band broadening.
- ✓ Superficially porous packing reduce volume of stagnant pools of mobile phase thus reducing band broadening.

HPLC...

- The packing used in modern HPLC consists of small, rigid particles having a narrow particle size.
- There are mainly three types of column packings.
 - (a) Porous, polymeric beads based on styrene–divinyl benzene copolymers. These are mainly used for ion exchange and size exclusion chromatography.

- Nowadays, these are replaced by silica based packings which are more efficient and mechanically more stable.

(b) Porous layer beds consisting of a thin shell of silica or modified silica on a spherical inert core.

(c) Totally porous silica particles with narrow particle size (diameter $<10\ \mu\text{m}$) range.

HPLC...

4. HPLC Columns...

Stationary phases...

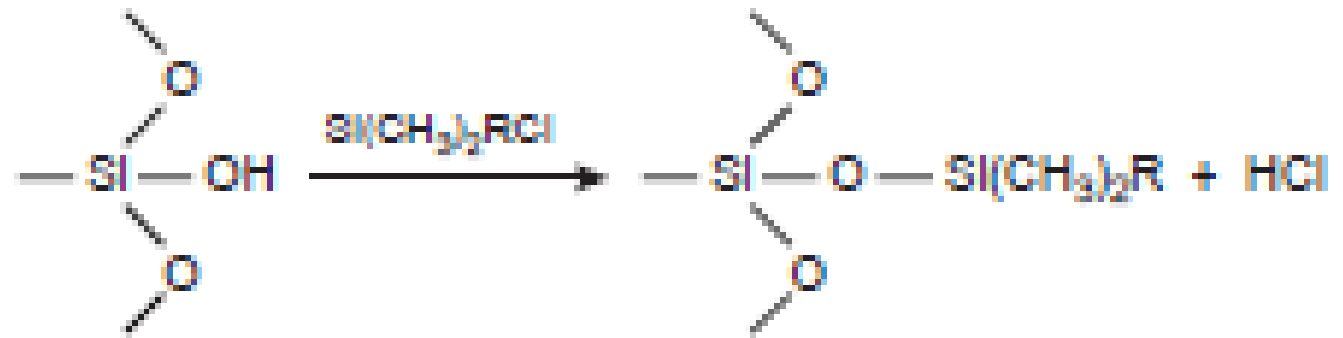
- **In liquid–liquid chromatography** the stationary phase is a liquid film coated on a packing material consisting porous silica particles.
- The stationary phase may be partially soluble in the mobile phase, causing it to “bleed” from the column.
- To prevent this loss of stationary phase, it is covalently bonded to the silica particles.
- **Bonded stationary phases are attached by reacting the silica particles** with an organ chlorosilane of the general form $\text{Si}(\text{CH}_3)_2\text{R}$, where **R** is an alkyl or substituted alkyl group.

HPLC...

4. HPLC Columns...

Stationary phases...

- The properties of a stationary phase are determined by the nature of the organosilane's alkyl group.
- If R is a polar functional group, then the stationary phase will be polar.
- Examples of polar stationary phases include those for which R contains a cyano ($-\text{C}_2\text{H}_4\text{CN}$), diol ($-\text{C}_3\text{H}_6\text{OCH}_2\text{CHOHCH}_2\text{OH}$), or amino ($-\text{C}_3\text{H}_6\text{NH}_2$) functional group.



HPLC...

4. HPLC Columns...

Stationary phases...

- The most common nonpolar stationary phases use an organochlorosilane for which the R group is an *n*-octyl (C_8) or *n*-octyldecyl (C_{18}) hydrocarbon chain.
- In **liquid–solid adsorption chromatography (LSC)** the **column packing also serves** as the stationary phase.
- In Tswett's original work the stationary phase was finely divided CaCO_3 , but modern columns employ porous 3–10-mm particles of silica or alumina.
- Since the stationary phase is polar, the mobile phase is usually a nonpolar or moderately polar solvent.

HPLC...

4. HPLC Columns...

Stationary phases...

- Typical mobile phases include hexane, isooctane, and methylene chloride
- The usual order of elution, from shorter to longer retention times (NP) , is

**olefins < aromatic hydrocarbons < ethers < esters, aldehydes,
ketones < alcohols, amines < amides < carboxylic acids**

HPLC...

5. Detectors

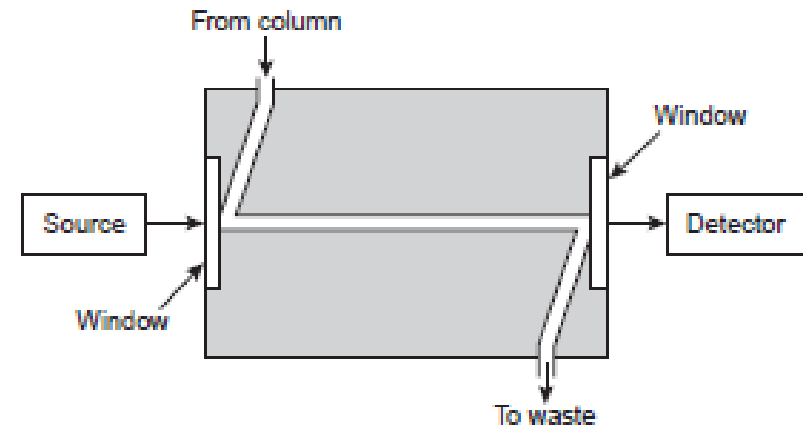
- Numerous detectors have been developed for use in monitoring HPLC separations
- Different methods of detection are available
 - **Spectroscopic methods**
 - UV-Visible spectroscopy
 - Fluorescence spectroscopy
 - Refractive index spectroscopy
 - Atomic spectroscopic techniques
 - **Electrochemical methods**
 - Conductometers
 - Potentiometers
 - Voltammeters
 - **Mass spectroscopy**

HPLC...

5. Detectors...

A. UV/Visible detector

- An ultraviolet light beam is directed through a flow cell and a sensor measures the light passing through the cell.
- If a compound elutes from the column that absorbs this light energy, it will change the amount of light energy falling on the sensor.
- The resulting change in this electrical signal is amplified and directed to a recorder or data system.



HPLC...

5. Detectors...

A. UV/Visible detector...

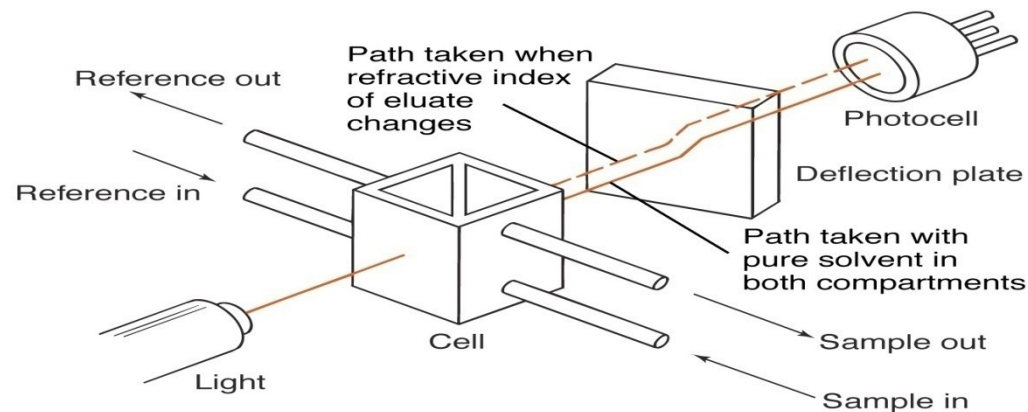
- Non-destructive, not-universal
 - not all compounds absorb light
 - Non-absorbing compounds can be analyzed by derivatizing them into light absorbing species.
 - This is done by mixing the effluent with chromogenic reagent in post column reactor.
- It has moderate sensitivity
- When using a UV/Vis detector, the resulting chromatogram is a plot of absorbance as a function of elution time.
- The flow cell has a volume of 1–10 μL and a path length of 0.2–1 cm.
- One limitation to using absorbance is that the mobile phase must not absorb strongly at the selected wavelength.

HPLC...

5. Detectors...

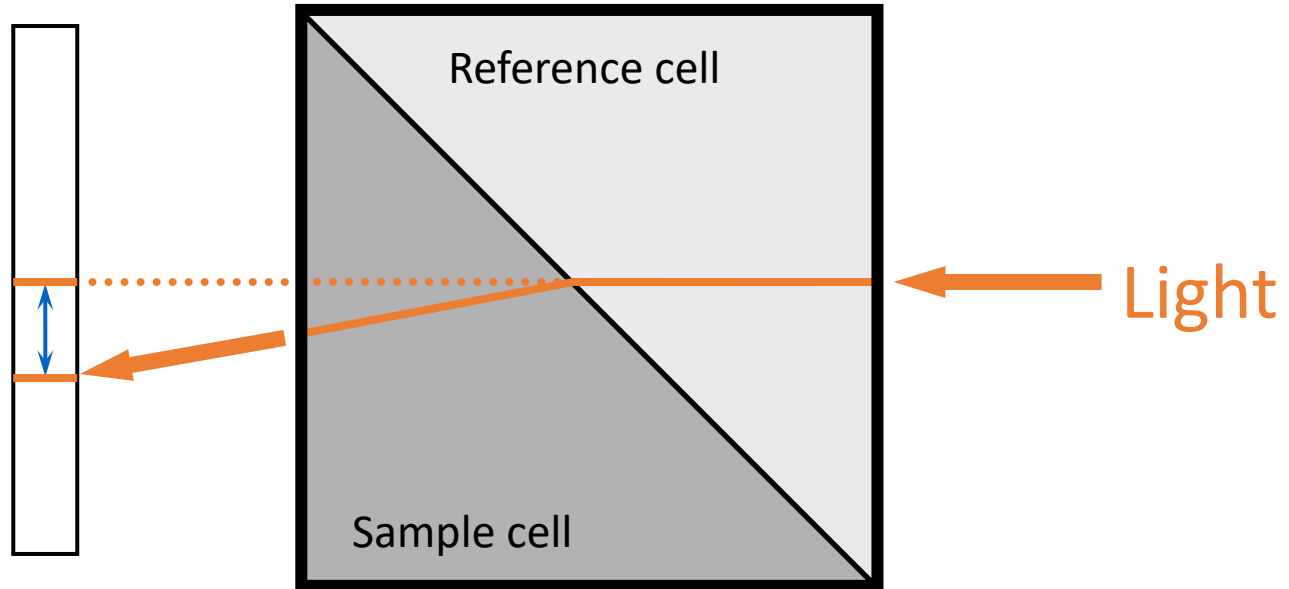
B. Refractive index (RI) detectors

- The ability of a compound or solvent to deflect light provides a way to detect it.
- Refractive index of a medium is the ratio of speed of light in a vacuum to its speed in the medium.
- These detectors measure the change in **refractive index** of the eluent as the solute passes through the sample cell.



HPLC...

Light-receiving unit



HPLC...

5. Detectors...

B. Refractive index (RI) detectors

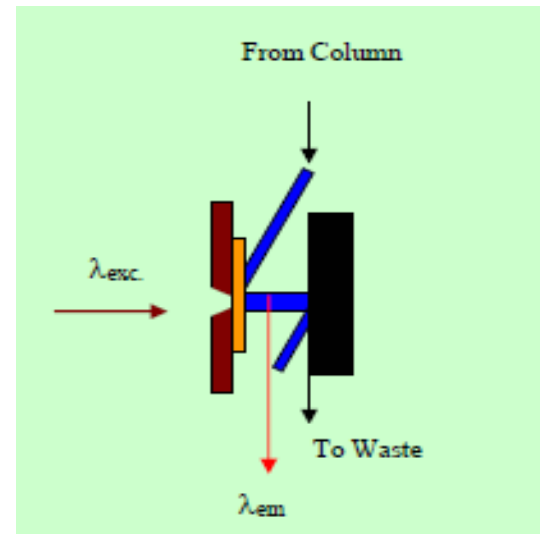
- The amount of deflection is proportional to concentration.
- The RI detector is considered to be a **universal detector** but it is **not very sensitive**.
- Sensitive to changes in pressure and temperature
- Useless in gradient elution due to change in the solvent index of refraction as the gradient is changed.
- Can not be used when the solvent has an index of refraction close to that of solutes.
- Not useful for trace analysis.

HPLC...

5. Detectors...

C. Fluorescence detectors

- Compared to UV-Vis detectors fluorescence detectors offer a higher sensitivity and selectivity that allows to quantify and identify compounds in complex matrices at extremely low concentration levels (trace level analysis).
- Fluorescence detectors sense only those substances that fluoresce



Parameters

Several factors affect efficiency in chromatographic techniques including:

- (a) Particle size
- (b) Flow rate
- (c) Thickness of stationary phase
- (d) Mobile-phase viscosity
- (e) Diffusion of solute in mobile and stationary phases
- (f) Column packing
- (g) Sample size (μg sample/g packing)

Particle size

- In the HPLC, the stationary phase attached to the outside of small spherical silica particles (very small beads).
- Particles from 5 μm size beads are the most commonly available.
- Smaller particles provide more surface area and better separations.
- But the pressure required for optimum linear velocity increases by the inverse of the particle diameter squared.
- This means that changing to particles that are half as big, keeping the size of the column the same, will double the performance, but increase the required pressure by a factor of 4.
- Larger particles are used in preparative HPLC (column diameters 5 cm up to >30 cm) and for non HPLC applications such as solid-phase extraction.

HPLC...

□ Applications of HPLC

- HPLC is routinely used for both **qualitative and quantitative analysis** of environmental, pharmaceutical, industrial forensic, clinical, and consumer product samples.
- Separation and analysis of non-volatile or thermally-unstable compounds
- Typical non-volatile compounds are:
 - Pharmaceuticals like aspirin, ibuprofen, or acetaminophen (Tylenol)
 - Salts like sodium chloride and potassium phosphate
 - Proteins like egg white or blood protein
 - Organic chemicals like polymers (e.g. polystyrene, polyethylene)
 - Heavy hydrocarbons like asphalt or motor oil
 - Thermally unstable compounds such as trinitrotoluene (TNT), enzymes

HPLC...

□ Applications of HPLC

➤ Preparation of Pure Compound(s)

- By collecting the chromatographic peaks at the exit of the detector, and concentrating the compound (analyte) by removing/evaporating the solvent,
- A pure substance can be prepared for later use (e.g. *organic synthesis, clinical studies, toxicology studies, etc.*).

